



例子：用ATF-34143设计2.4GHz低噪声放大器。

放大器的设计步骤总结如下：

步骤一：在频率、增益、噪声指标条件下选择器件，
得到偏置条件下器件的 $[S]$ 参数。

步骤二：满足稳定判据，则进入步骤三；不满足判据，
在圆图中画出稳定区，必要时画出等增益圆
和等噪声圆。

步骤三：确定输入输出反射系数 Γ_{in} 及 Γ_{out} 。

步骤四：设计输入输出匹配网络。



增加ATF34143模型到模型库中

① 打开ADS，菜单【DesignKits】→【Unzip Design Kit】

本地磁盘 (C:) > Program Files > Keysight > ADS2015_01 > oalibs > componentLib

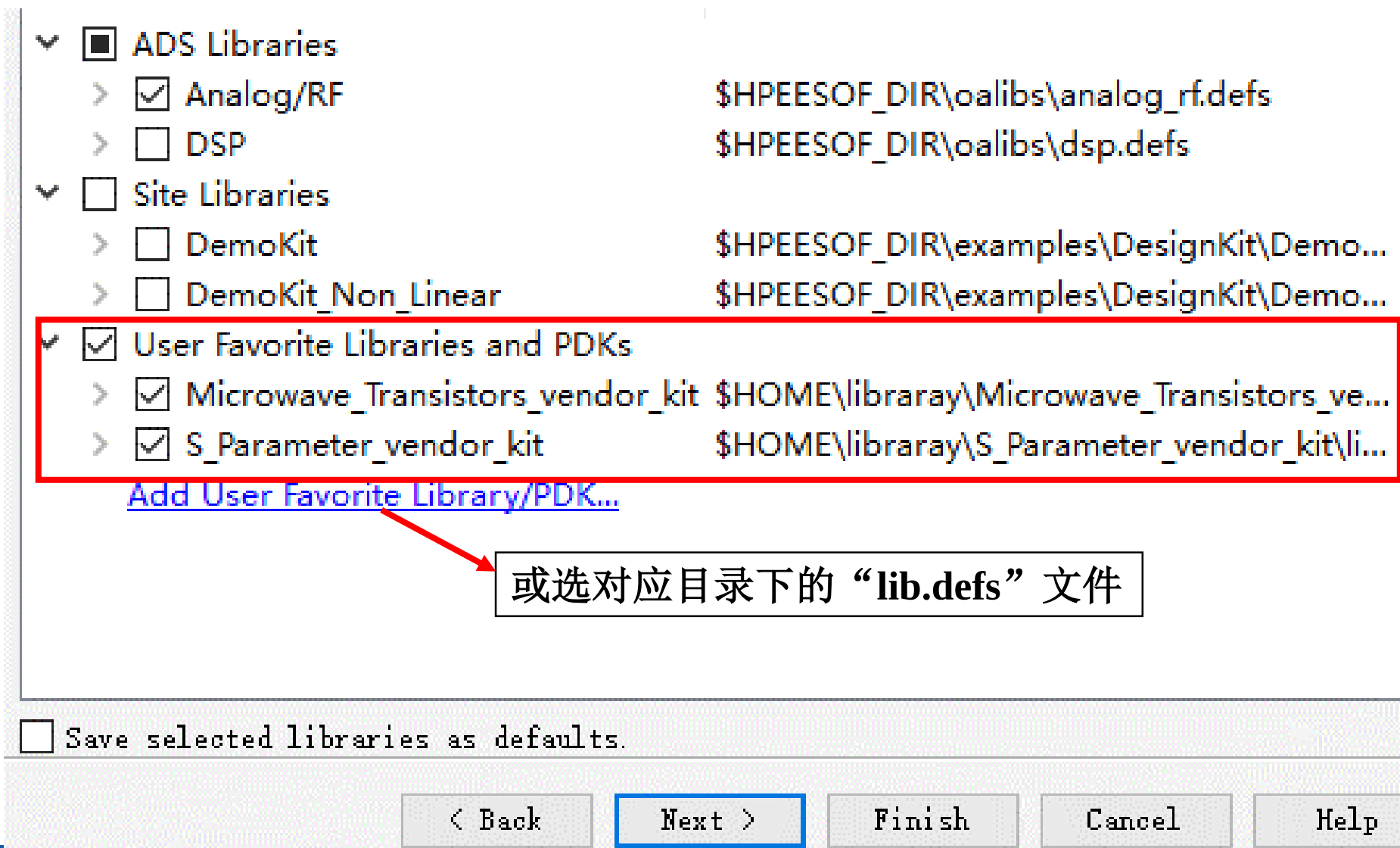
名称	修改日期	类型
Analog_Parts_vendor_kit.7z	2014/11/8 12:53	WinRAR 压缩文件
HFDiode_vendor_kit.7z	2014/11/8 12:53	WinRAR 压缩文件
Meas_SMT_Passive_vendor_kit.7z	2014/11/8 12:53	WinRAR 压缩文件
Microwave_Transistors_vendor_kit.7z	2014/11/8 12:53	WinRAR 压缩文件
RF_Passive_SMT_vendor_kit.7z	2014/11/8 12:53	WinRAR 压缩文件
RF_Transistors_vendor_kit.7z	2014/11/8 12:53	WinRAR 压缩文件
S_Parameter_vendor_kit.7z	2014/11/8 12:53	WinRAR 压缩文件

② 释放到目标文件夹（如C:\Users\DELL\library）



1、三极管直流工作点分析及偏置电路设计

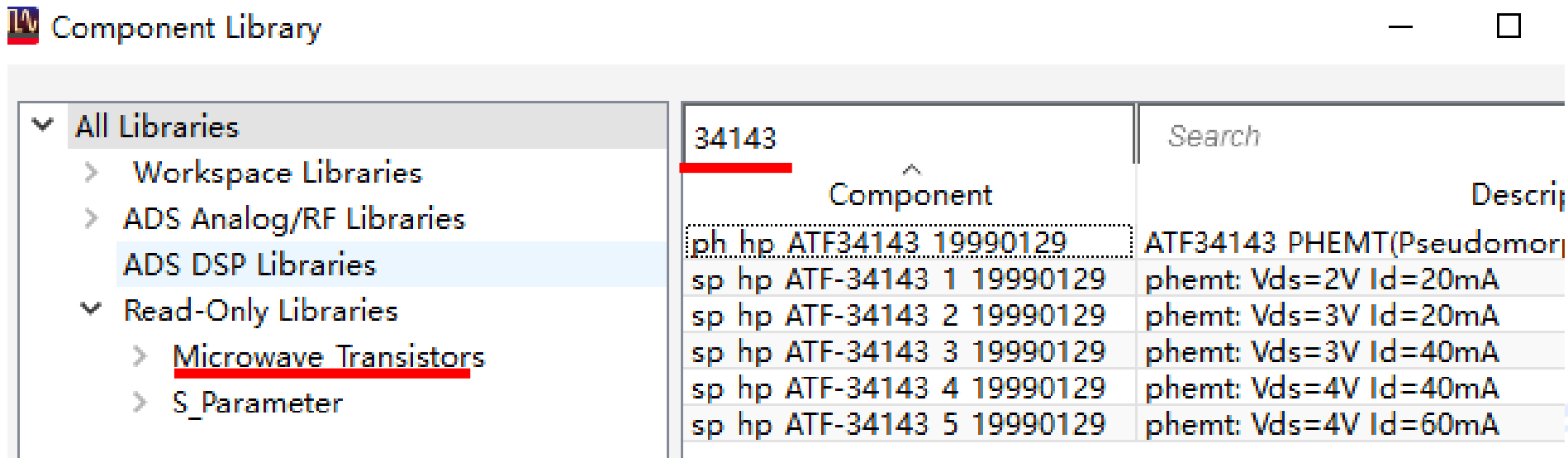
➤ 建新工程，工程向导中选择新加的模式库





➤ 建立新设计

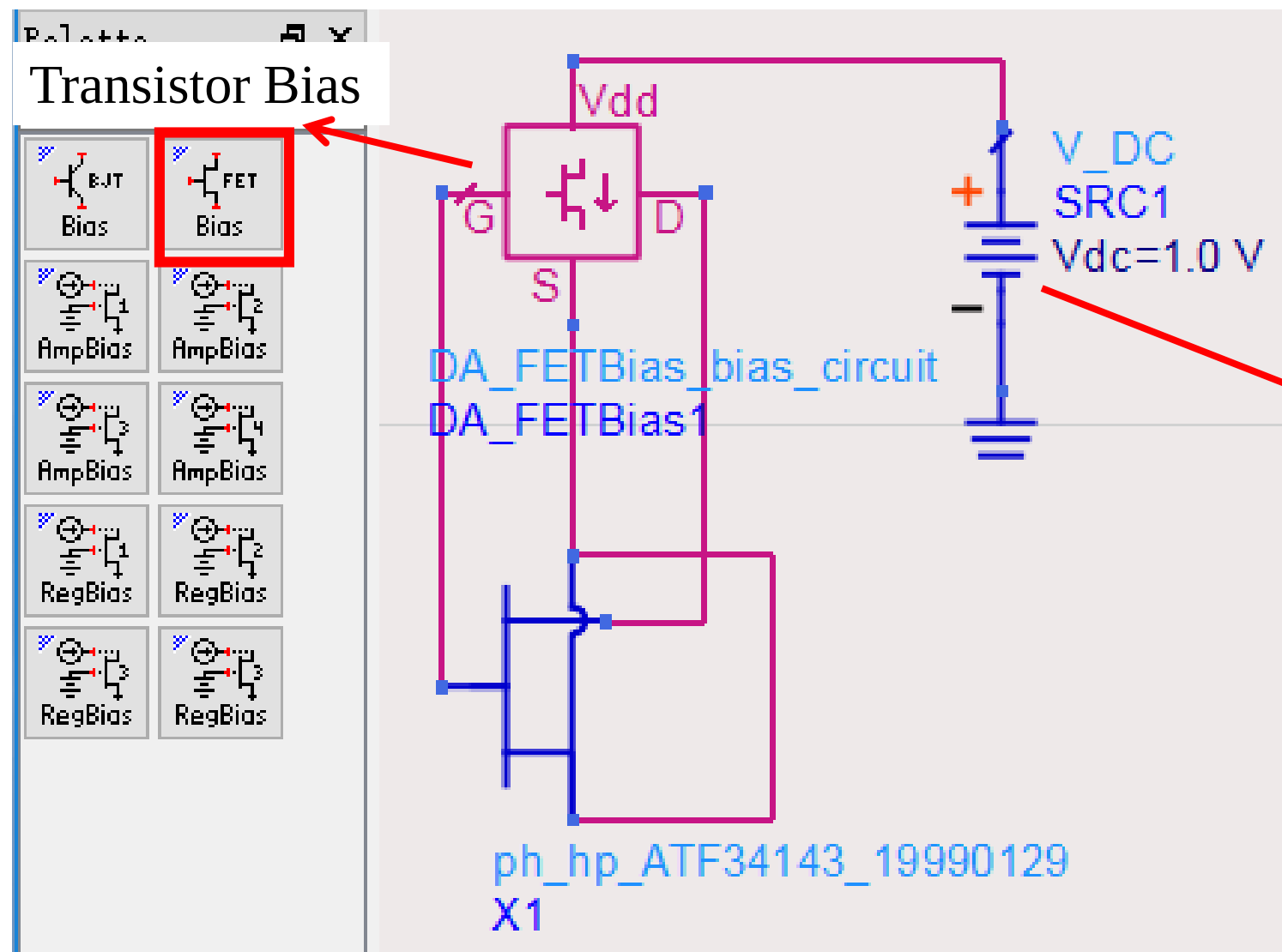
➤ 查询需要的晶体管：点击 ，打开元件库



- sp为开头的是S参数模型
- 选择ph开头的模型，将晶体管拖入原理图窗口，放入晶体管



完成原理图



Sources-Freq Domain





➤ 执行命令【DesignGuide】→【Amplifer】



Amplifier:1

- > DC and Bias Point Simulations
- > S-Parameter Simulations
- > 1-Tone Nonlinear Simulations
- > 2-Tone Nonlinear Simulations
- > Power Amplifier Examples - By Class of Operation
- > Amplifier Statistical Design
- > Lumped 2-Element Z-Y Matching Networks
- > Lumped Multi-Element Z-Y Matching Networks
- ▼ Tools
 - Transistor Bias Utility
 - Smith Chart Utility
 - Matching Utility
 - Display Simulation Results
 - Amplifier DesignGuide Documentation
 - About Amplifier DesignGuide





Current Schematic

bias_circuit [LNA_lib:bias_circuit:schematic] ▼

SmartComponent

DA_FETBias1

Current Design

schematic

SmartComponent Capability

Design, Simulate, Extract Parameters

Overview

Resistive Networks

Active Networks

Bias Settings

Vdd

6

V ▼

Device Type

NFET ▼

Vds

5

V ▼

Id

10

mA ▼

☐ Include RF ChokesBias Point Selection

Device Parameters

☒ Automatically Extract Device Parameters☒ Device Parameters

Vt

-1

V ▼

K (A/V²)

1e-2

☐ Device Bias Point

Vgs

fV ▼

Helps

NFET

Vdd and Vds must be positive.

PFET

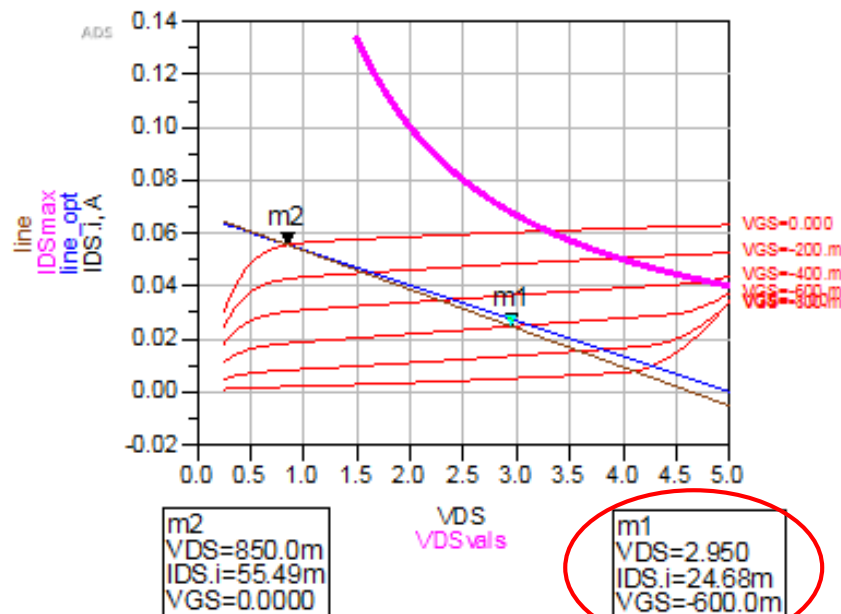
Vdd and Vds must be negative.





选择最优偏置点

Device IV Curves, Load Lines,
and Maximum DC Dissipation Curve



Follow these steps:

- 1) Move marker m2 to the knee of the I-V curve. This sets the maximum drain current during AC operation.
- 2) Specify maximum allowed VDS, VDSmax. The optimal bias point values are determined from the load line between marker m2 and the (IDS=0, VDS=VDSmax) point.
- 3) Specify maximum allowed DC power dissipation, PDmax, in Watts.
- 4) Position marker m1 at some other bias point, if desired. (Must be less than VDSmax.)
- 5) DC power consumption, average output power in linear operation, DC-to-RF efficiency at marker m1 bias point are all calculated.

Equations
are on the
"Equations"

Eqn VDSmax=5

Eqn PDmax=.2

Marker m1 bias point values, (Assuming Class A, AC current limited to marker m2 value and AC voltage no higher than VDSmax.)

Output Power Watts dBm		DC-to-RF Efficiency, %
25.30 m	14.03	34.75
Rload		DC Power Consumption
68.157		72.82 m

Optimal Class A bias point values.

Output Power
at Optimal Bias
Watts dBm

28.79 m

14.59

DC-to-RF
Efficiency, %

35.47

Optimal
VDS

2.925

Rload at
Optimal Bias

74.783

DC Power Consumption
at Optimal Bias

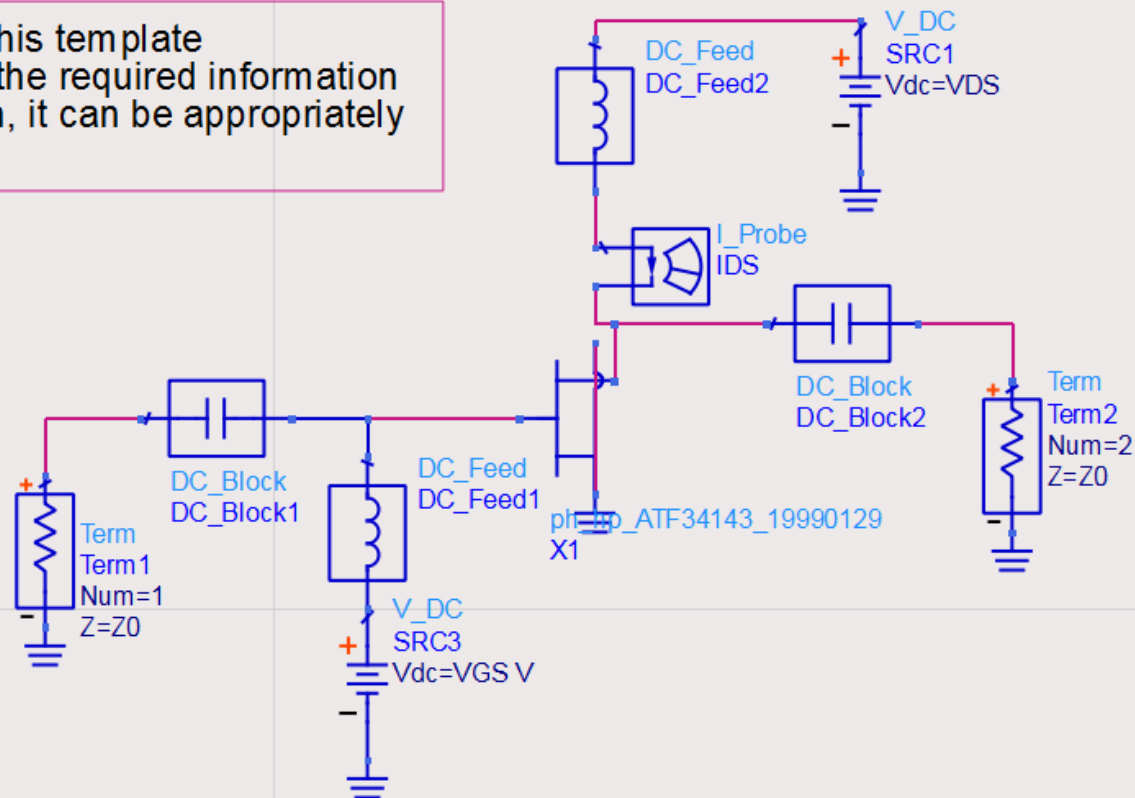
81.16 m

Optimal
IDS

27.75 m



In the event that this template does not provide the required information for the application, it can be appropriately modified.



Var
Eon

Var

```
VAR  
VAR3  
Z0=50  
VGSmin=-1  
VGSmax=0  
VGSstep=0.2  
VDSmin=0.25 V  
VDSmax=5 V  
VDSstep=0.1 V
```

Set characteristic impedance, and gate and drain voltage sweep limits as needed.



➤ 重新设置偏置参数

Overview

Resistive Networks

Active Networks

Bias Settings

Vdd

6

V

Device Type

NFET

Vds

3

V

Id

20

mA

☐ Include RF Chokes

Bias Point Selection

Device Parameters

☒ Automatically Extract Device Parameters

☒ Device Parameters

Vt

-1

V

☐ Device Bias Point

Vgs

fV

Helps

NFET

Vdd and Vds must be positive.

PFET

Vdd and Vds must be negative.

$I_d = K \cdot (V_{gs} - V_t)^2$

Design

Help



Bias Network Selection:1

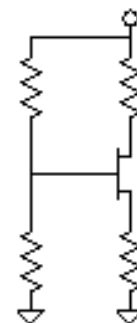
Use arrows to scroll and select desired network topology.

Fraction of DC Power Consumed in Gate Resistors

0.25 (0.000 < PR < 1.000)

Ratio of Source to Supply Voltage

0.111 0.111 < VR < 0.5



β of 3 networks

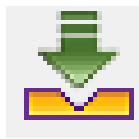
OK

Cancel

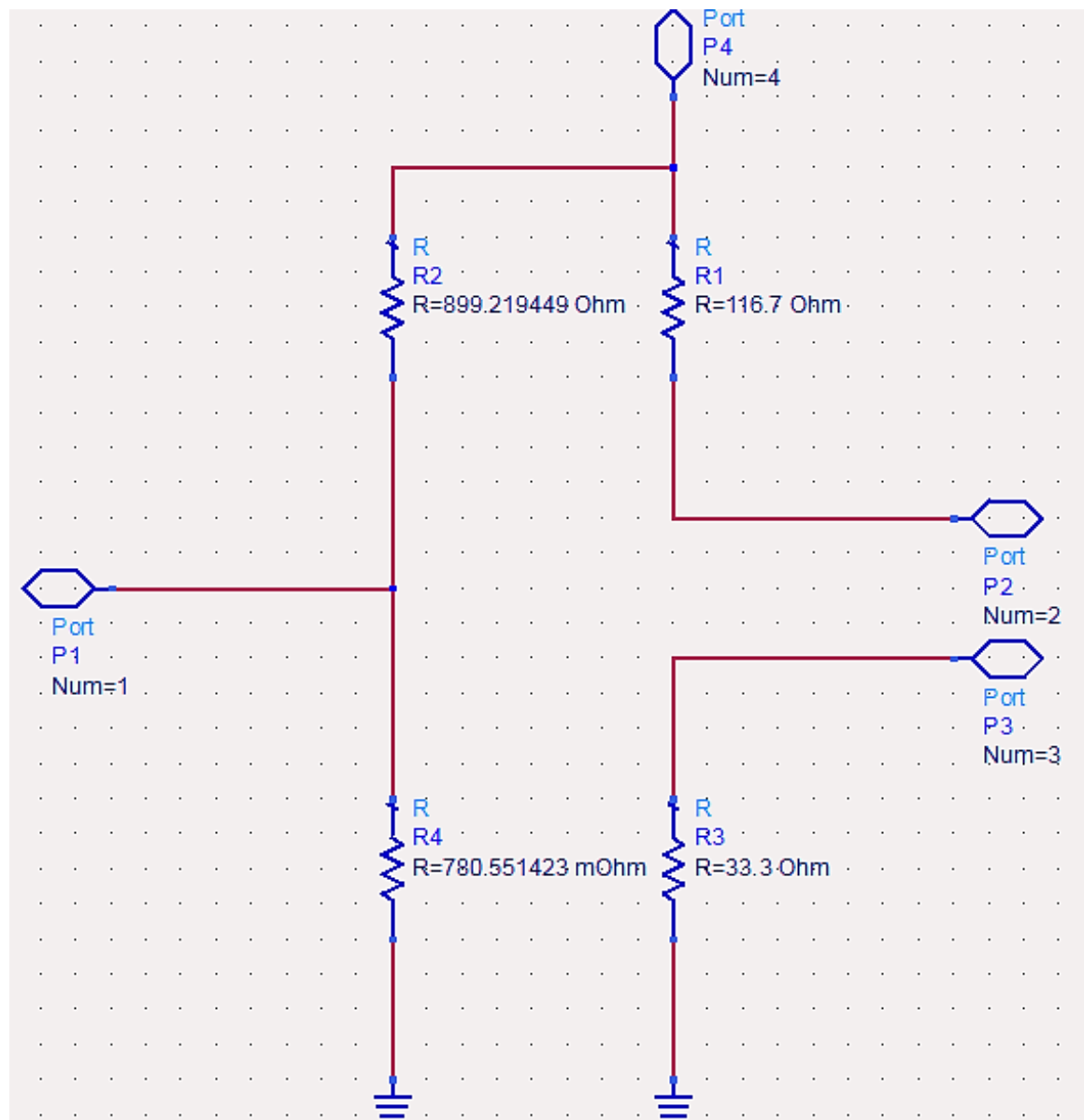


➤ 在原理图中单击

➤ 单击 返回



查看生成的偏置电路

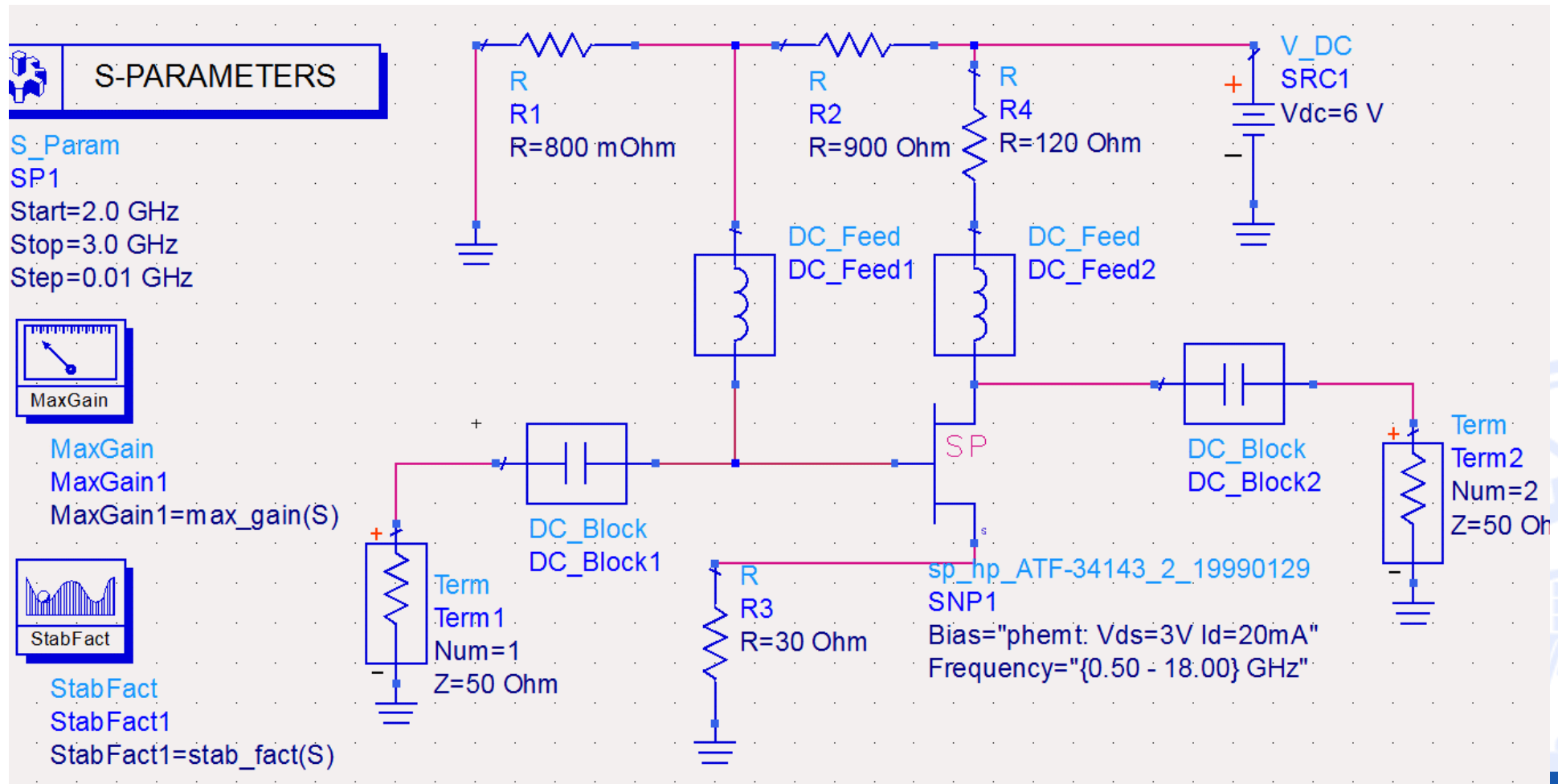


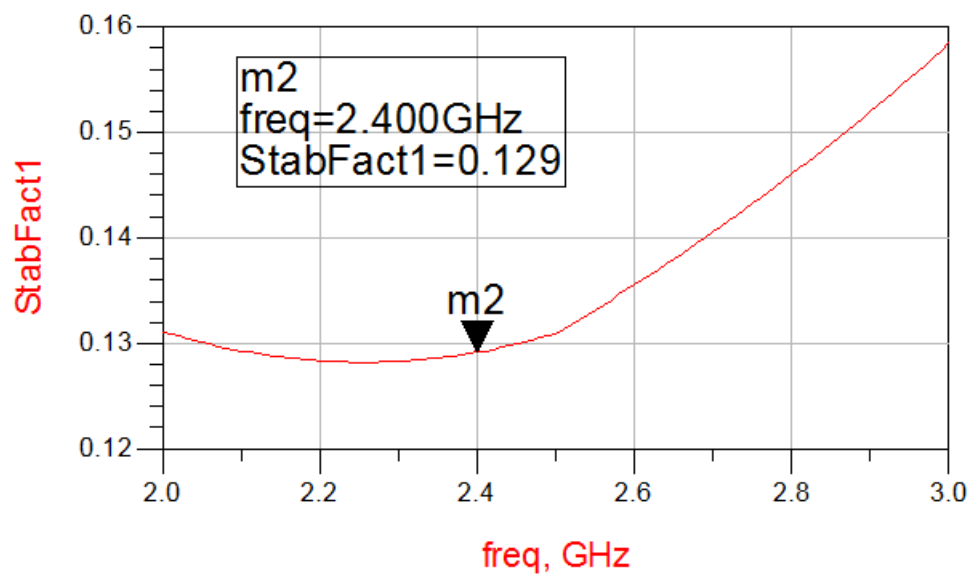
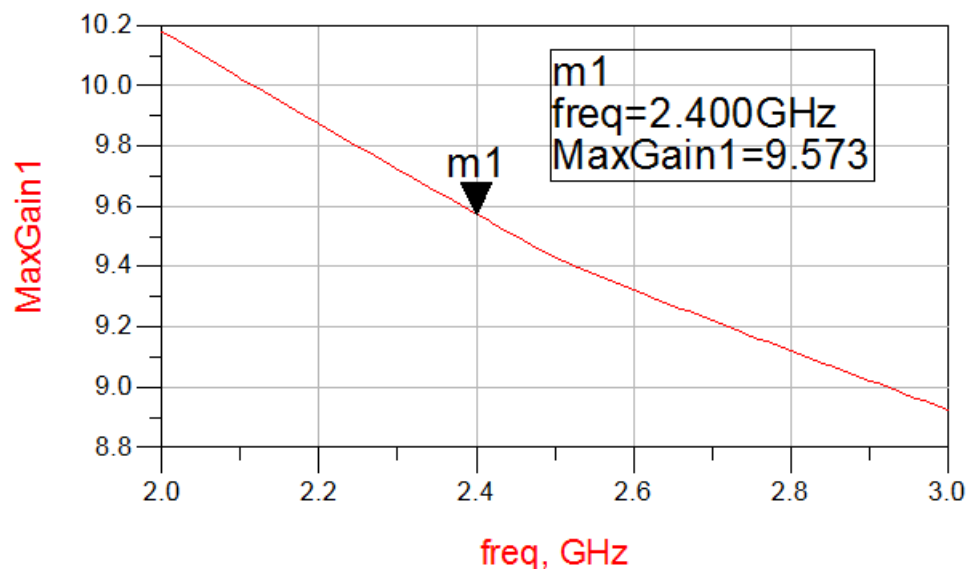


2、稳定性分析，查看稳定系数曲线

①新建一设计 名为sp_model，加入($V_{ds}=3V$ $I_d=20mA$)模型的晶体管

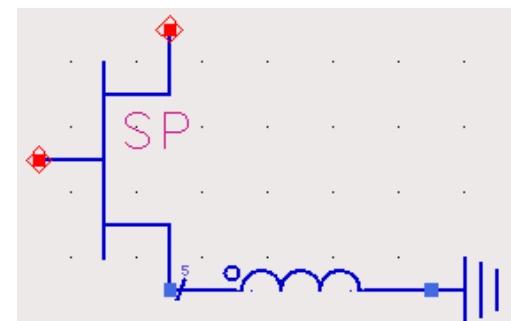
➤ 建立如下原理图插入稳定系数控件、最大增益控件→仿真



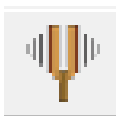


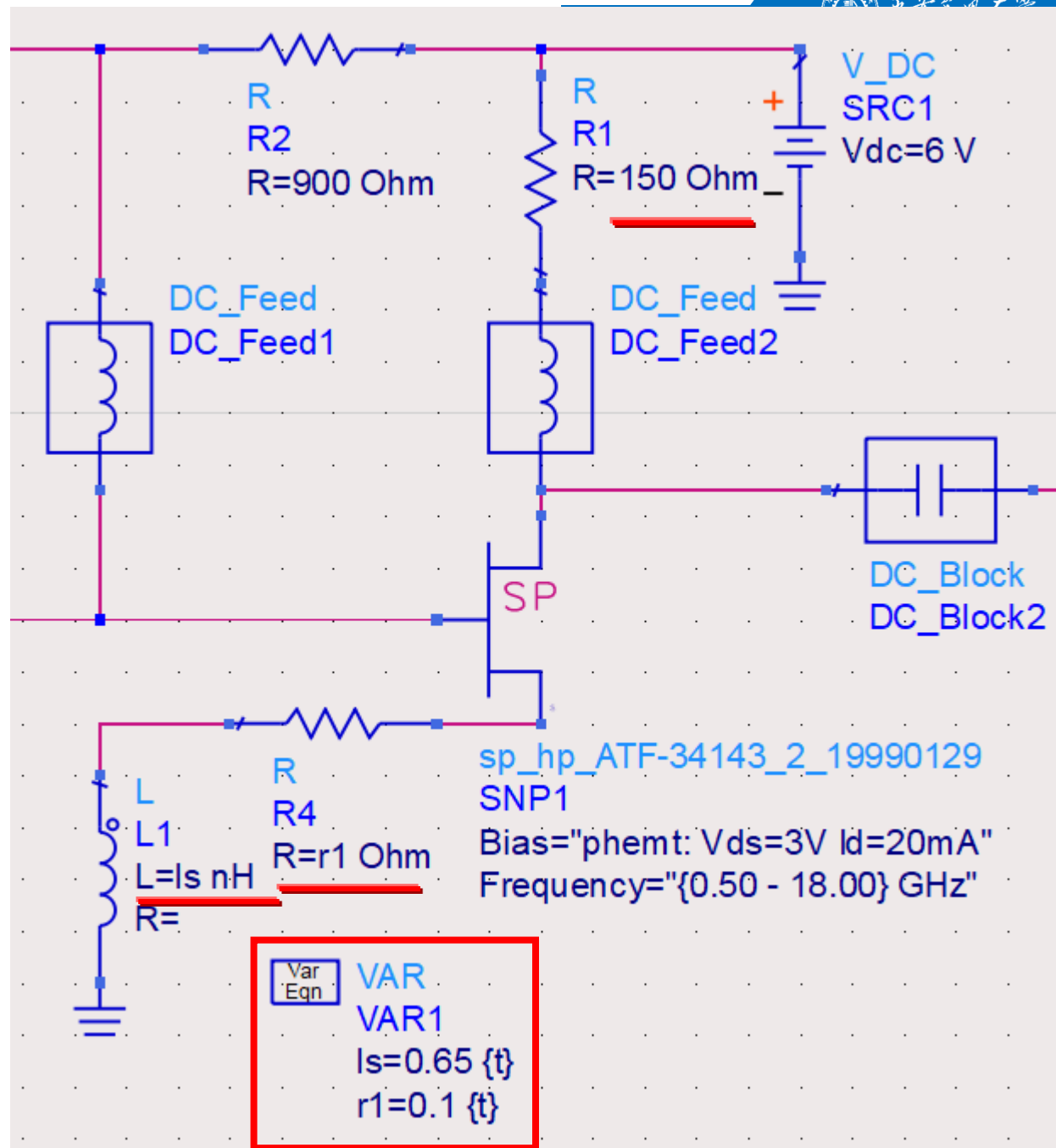
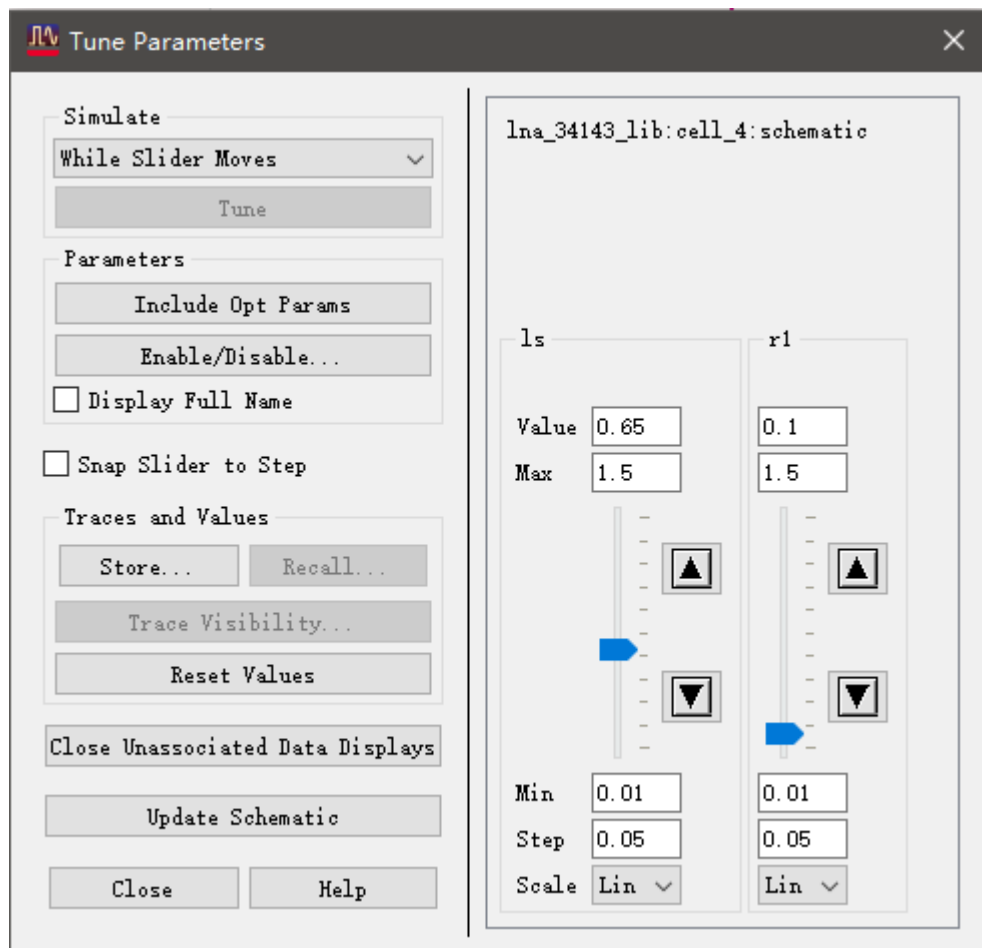
$$K < 1$$

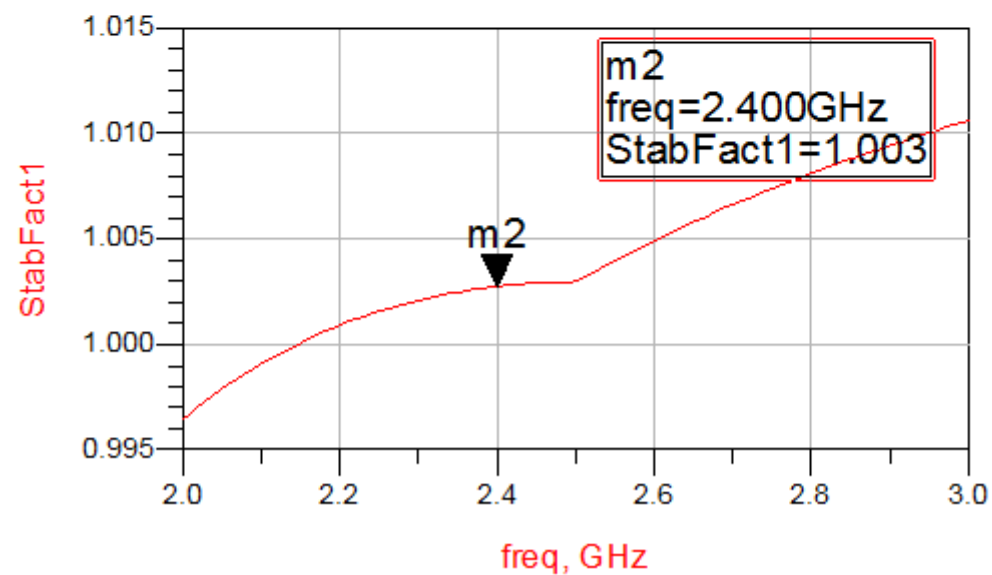
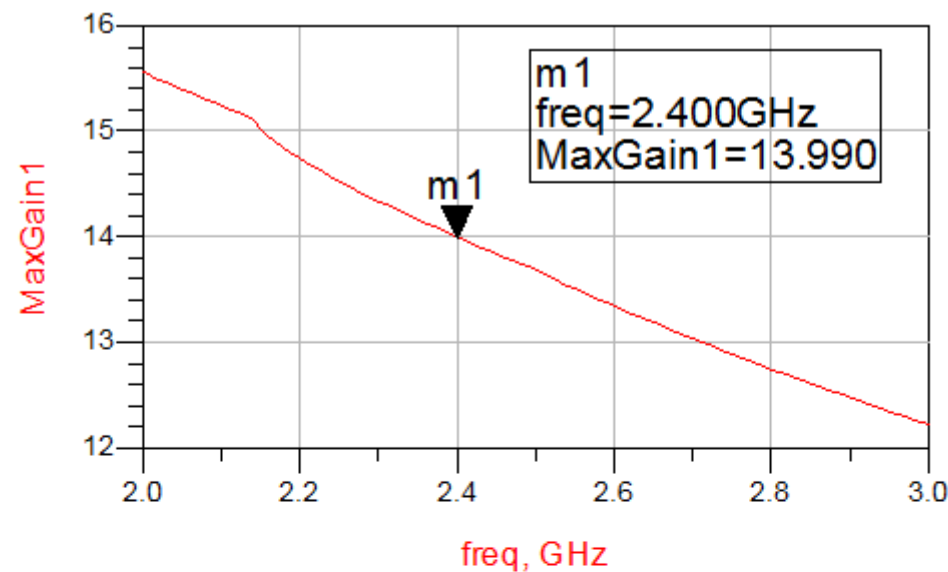
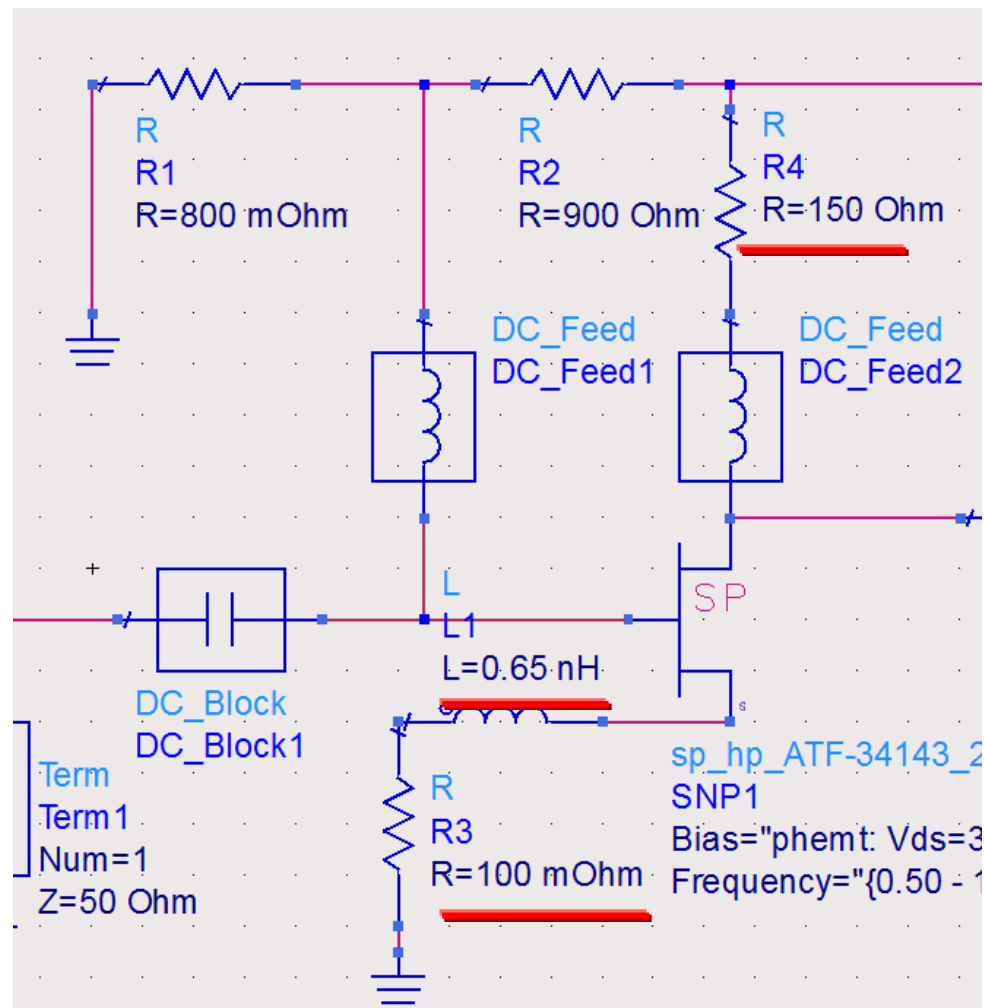
解决办法：
在源级加小电感抑制负反馈





- 源级加小电感和小电阻，设置变量
- 选择  调节参数，使其在工作频率范围内稳定

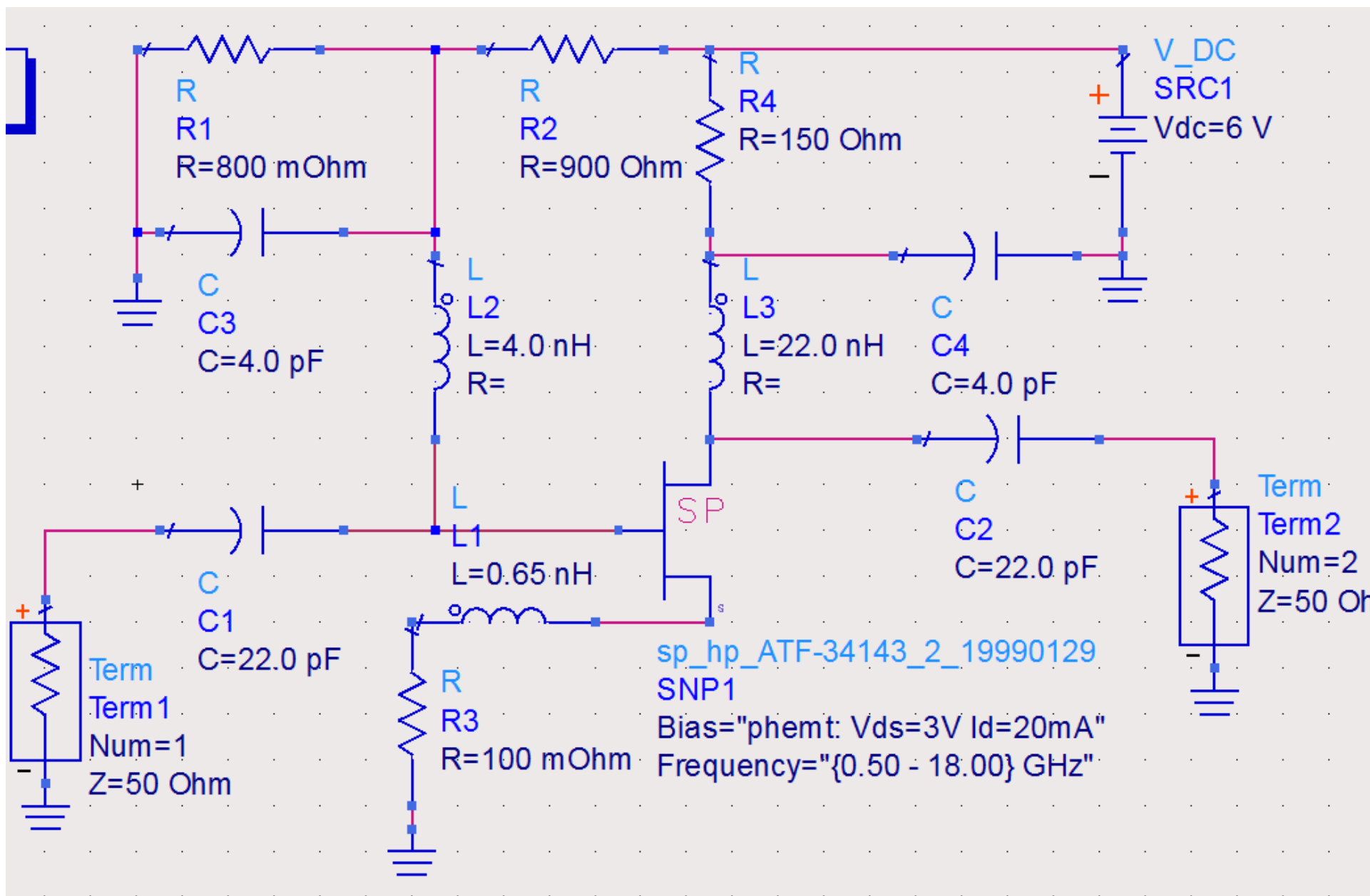




调整后，增益、稳定性都变好了

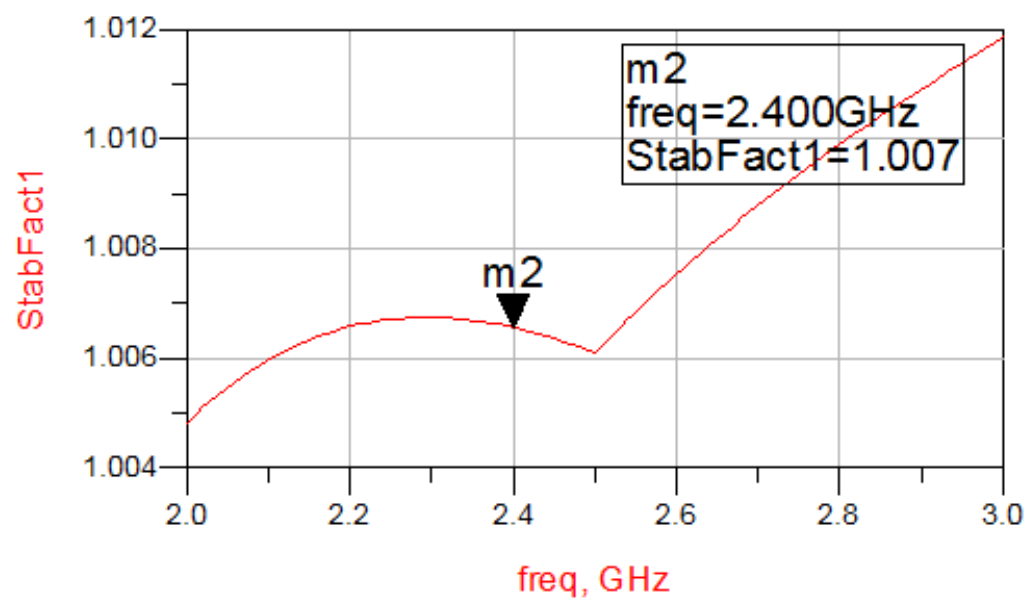
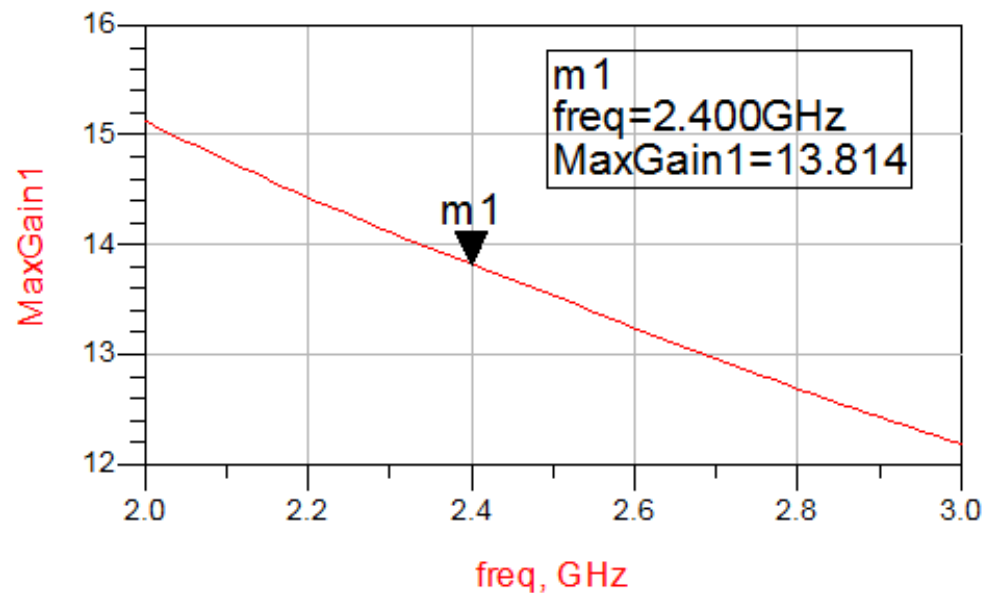


➤ 替换理想扼流电感和隔直电容；增加旁路电容





稳定性和增益依然很好

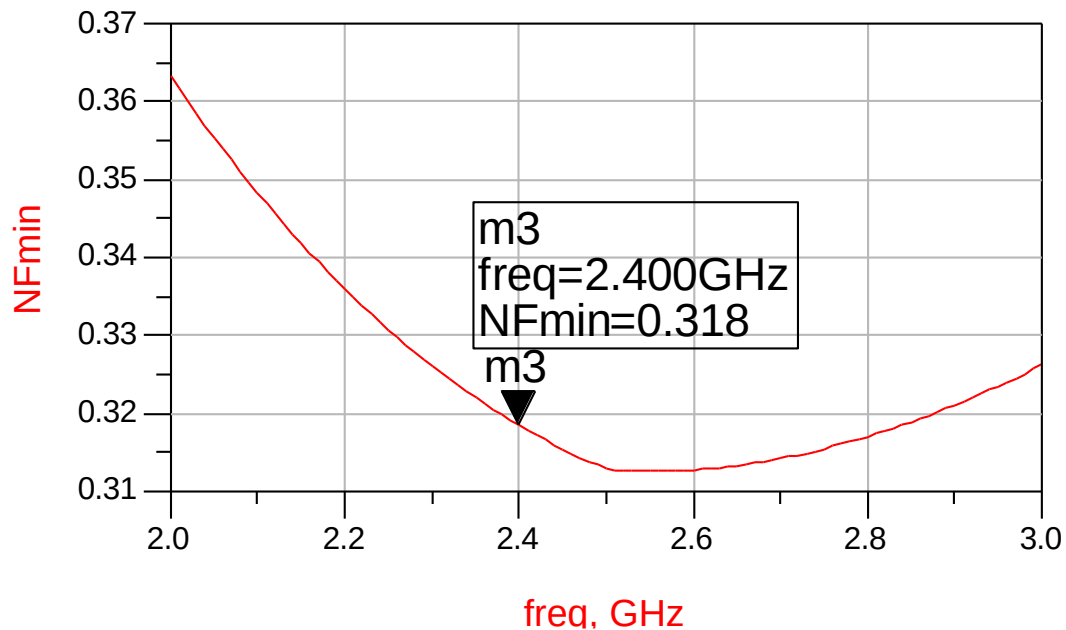




3、输入网络设计

➤ 加入噪声系数的仿真

➤ 双击  S-PARAMETERS



S_Param Instance Name
SP1

Frequency Parameters **Noise** Output

☒ Calculate noise

Noise input port

Noise output port

Noise contributors
Mode **Off**

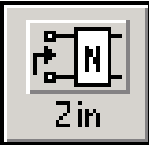
Dynamic range to display dB

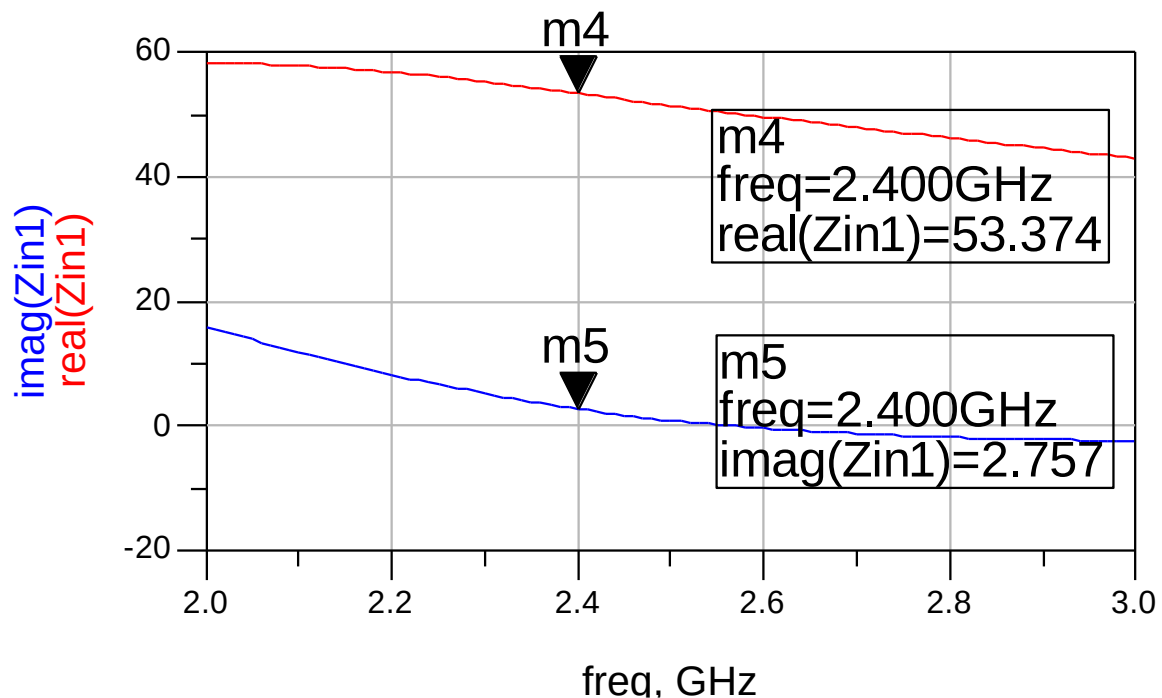
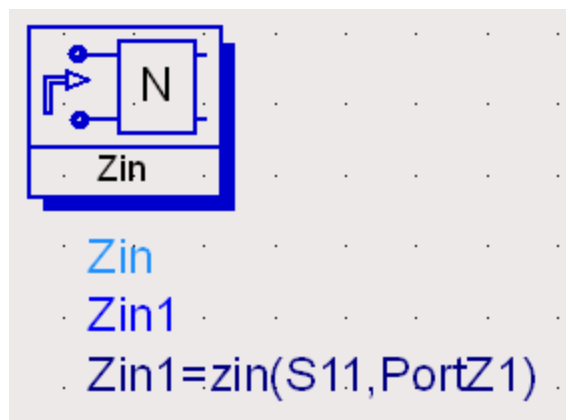
Bandwidth 1.0 Hz

OK Apply Cancel Help





- 点击元件库[Simulation-S_param]中的  ，放置输入阻抗测试控件→仿真



2.4GHz点的输入阻抗为： $Z_{in}=(53.374, 2.757)$



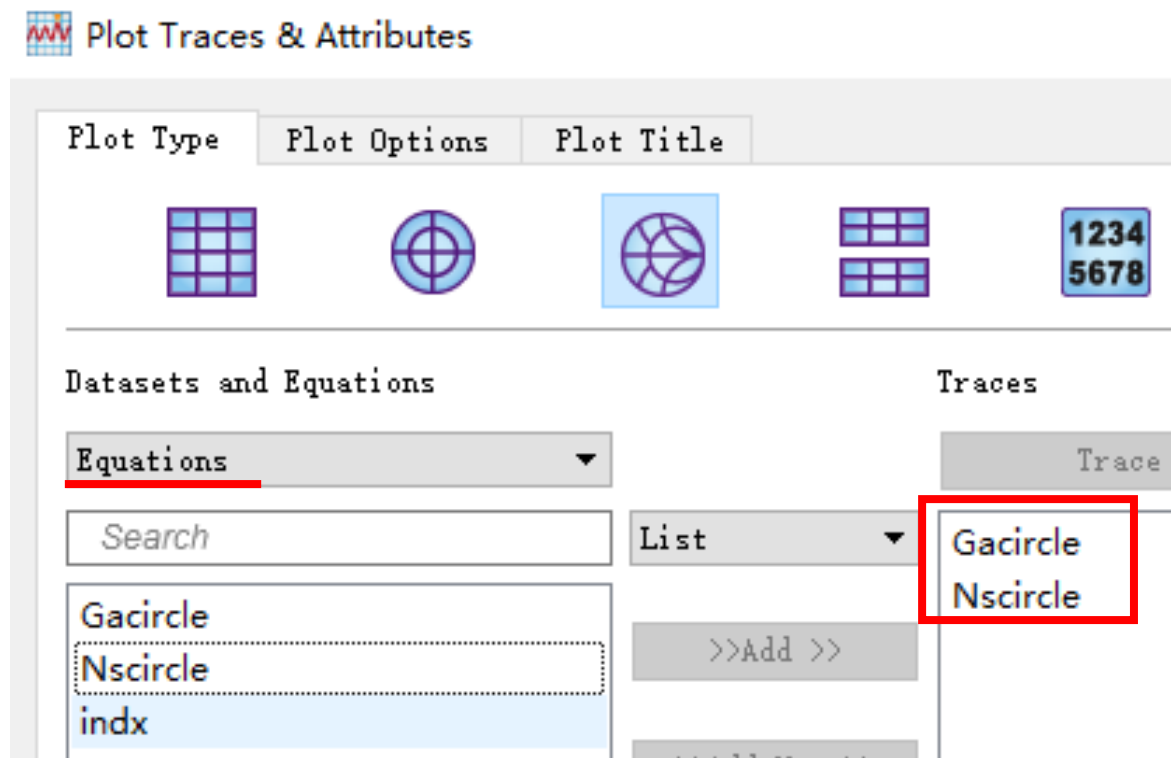
➤ 在数据窗口点击  添加下面内容

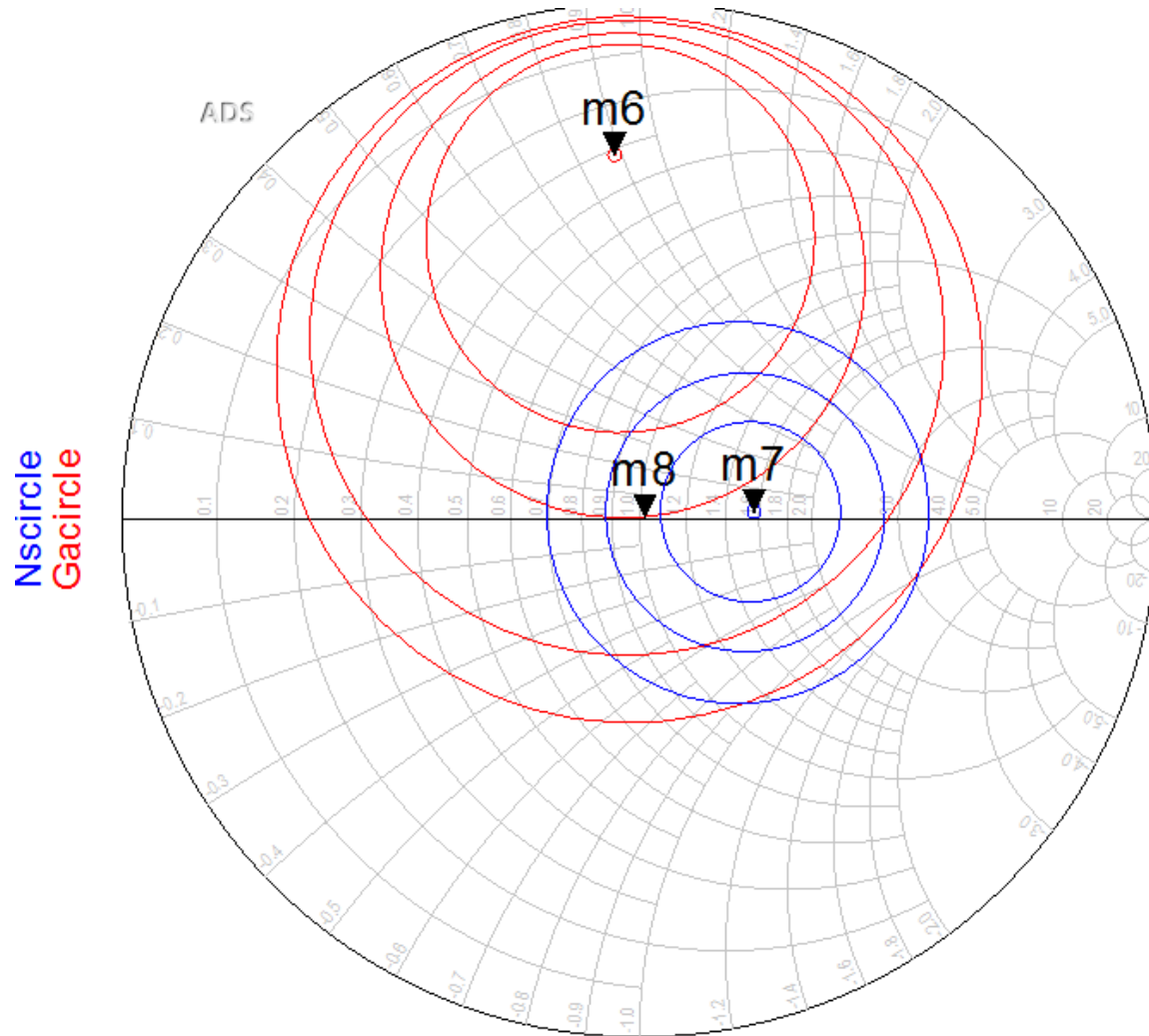
Eqn `indx=find_index(SP.freq,indep(m3))`

Eqn `Nscircle=ns_circle(NFmin[indx]+{0,0.02,0.05,0.1},NFmin[indx],Sopt[indx],Rn[indx]/50,51)`

Eqn `Gacircle=ga_circle(S[indx],MaxGain1[indx]-{0,0.5,0.8,1.5,2},51)`

➤ 在数据窗口点击





m6
indep(m6)=51
Gacircle=0.708 / 93.876
gain=13.814
impedance = $Z_0 * (0.313 + j0.884)$

m7
indep(m7)=51
Nscircle=0.222 / 4.104
ns figure=0.318
impedance = $Z_0 * (1.568 + j0.052)$

m8
indep(m8)=39
Gacircle=0.013 / 26.959
gain=13.014
impedance = $Z_0 * (1.023 + j0.012)$

最小噪声阻抗: $Z_0 \times (1.568 + j0.052) = 78.4 + j \times 2.6$

均衡阻抗: $Z_0 \times (1.023 + j0.012) = 51.15 + j \times 0.6$

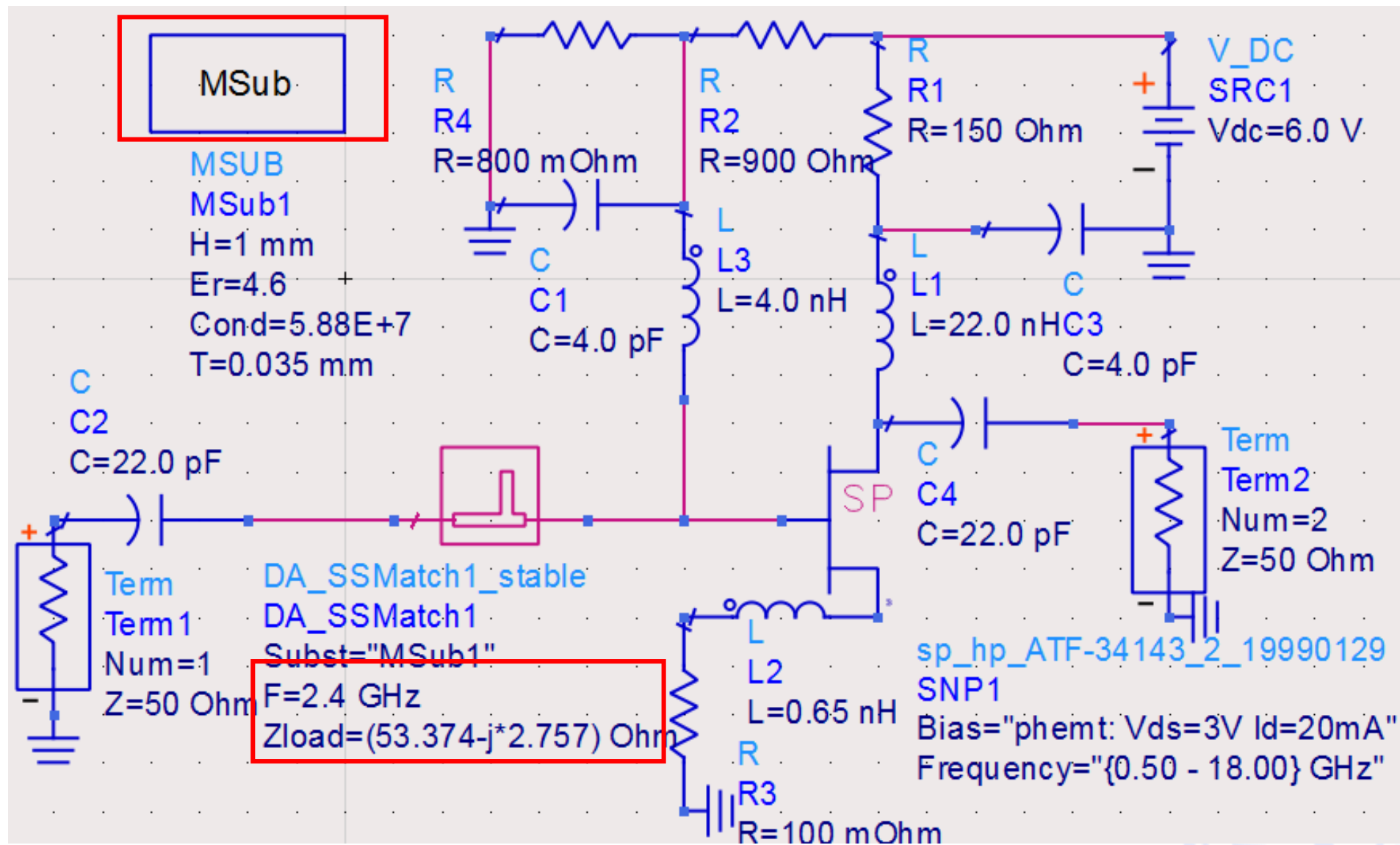
最大增益阻抗: $Z_0 \times (0.313 + j0.884) = 15.65 + j \times 44.2$

最小反射阻抗: $53.374 + j \times 2.757$





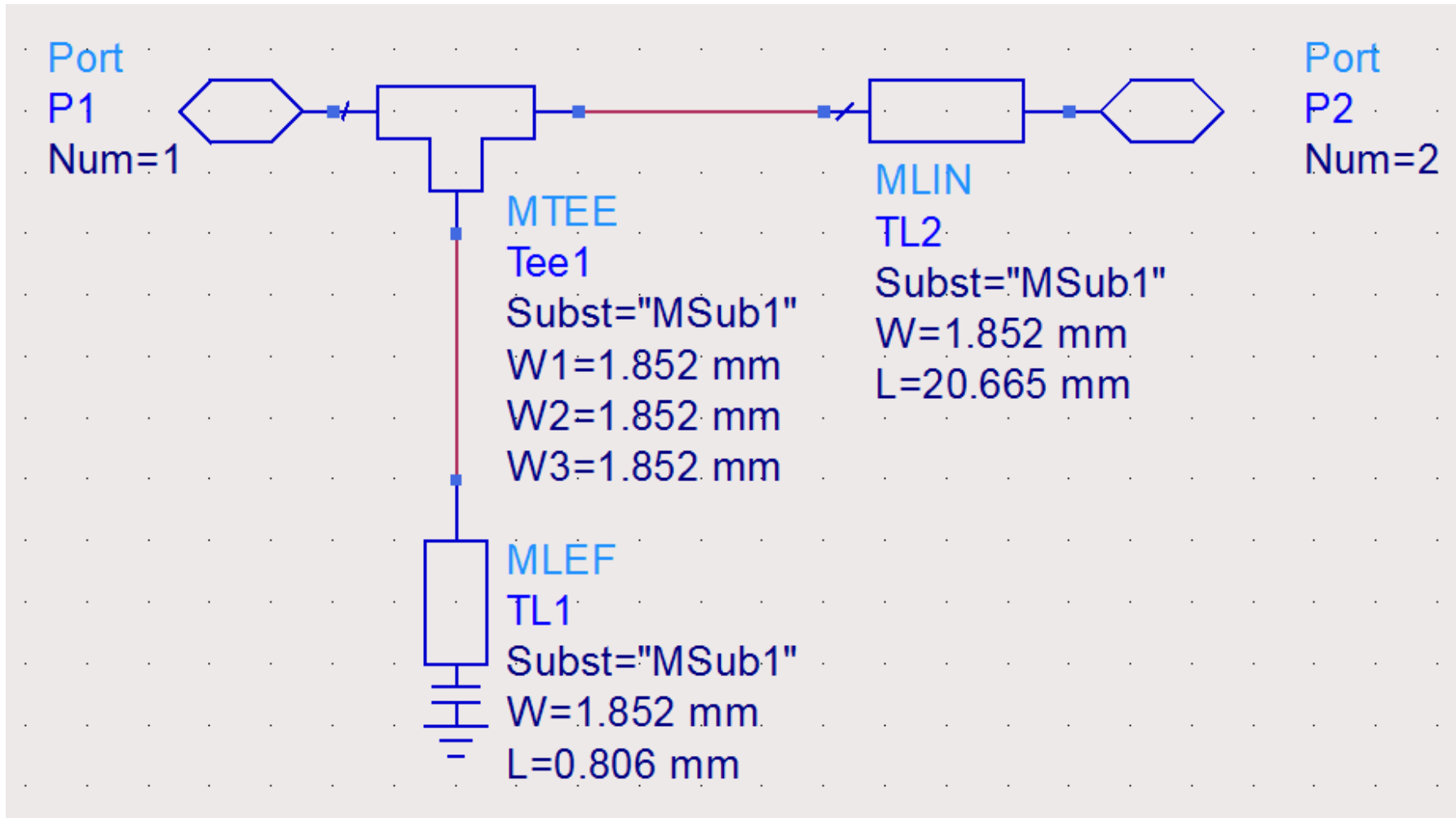
➤ 选择 “Passive Circuit DG-Microstrip Circuits” 元件库





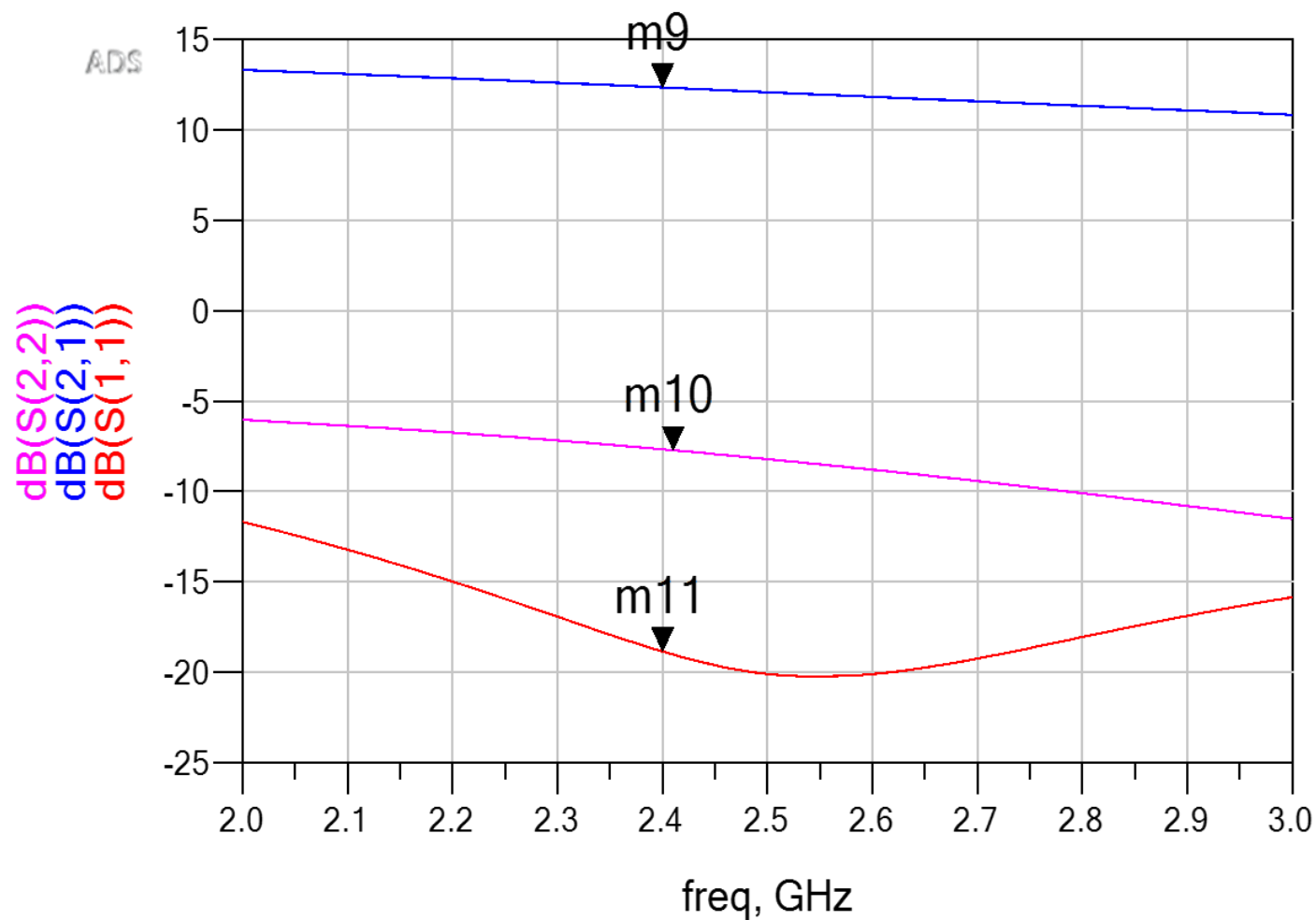
➤ 【DesignGuide】→【Passive Circuit】→【Passive Circuit Control Window】→自动设计

➤ 在原理图点击 ，查看生成的匹配网络。点击  返回





仿真结果

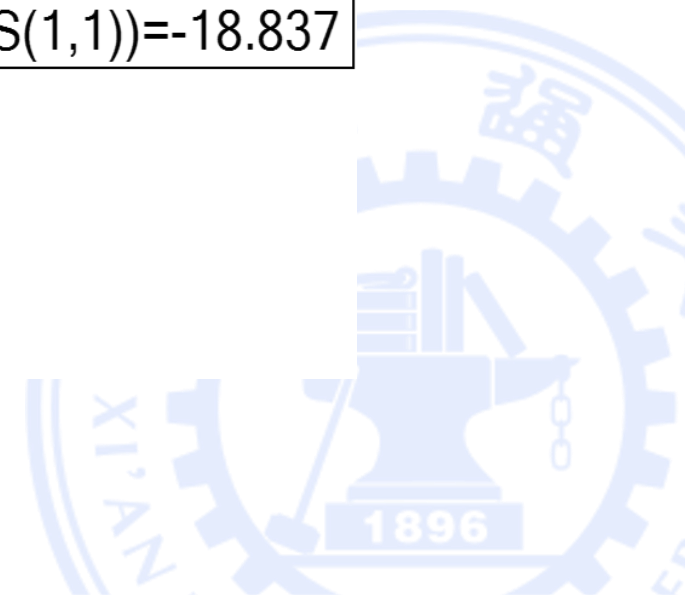


m9
freq=2.400GHz
dB(S(2,1))=12.321

m10
freq=2.410GHz
dB(S(2,2))=-7.721

m11
freq=2.400GHz
dB(S(1,1))=-18.837

输入端匹配良好，但输出端匹配不好





4、输出网络设计

➤ 放置分支线，并设置50Ohm线宽

➤ 设置TL1、TL2、TL3的支线长度取值（2—40mm）

Instance name: TL2

Select Parameter

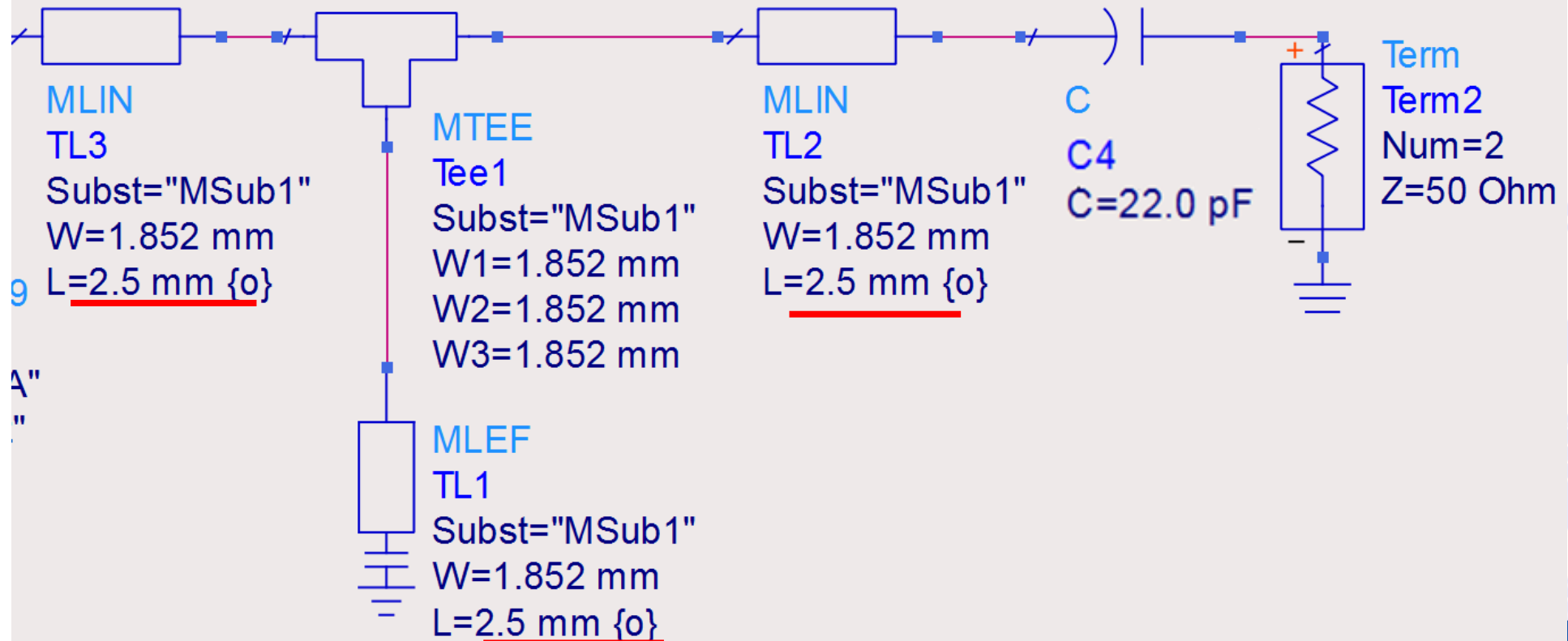
Subst="MSub1"

W=1.852 mm

L=2.5 mm opt{ 2 mm to 40 mm }

Wall1=2.5e+028 mm

Wall2=2.5e+028 mm



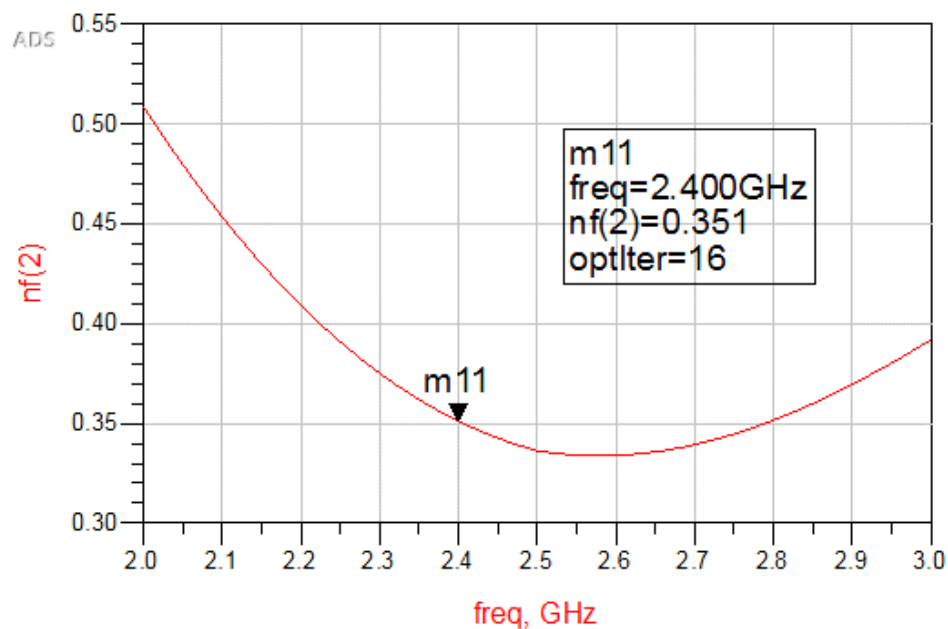
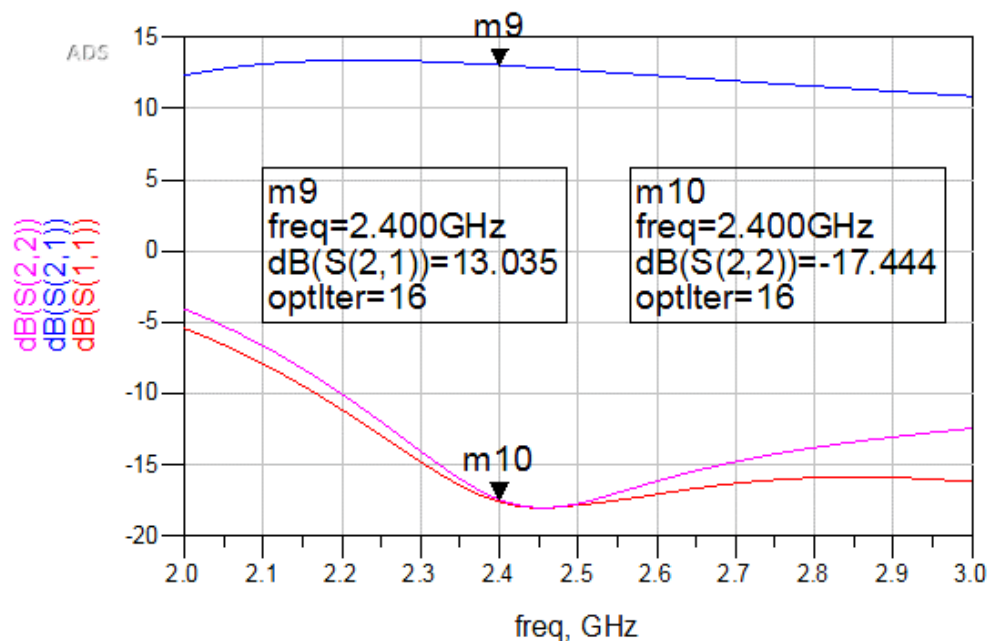
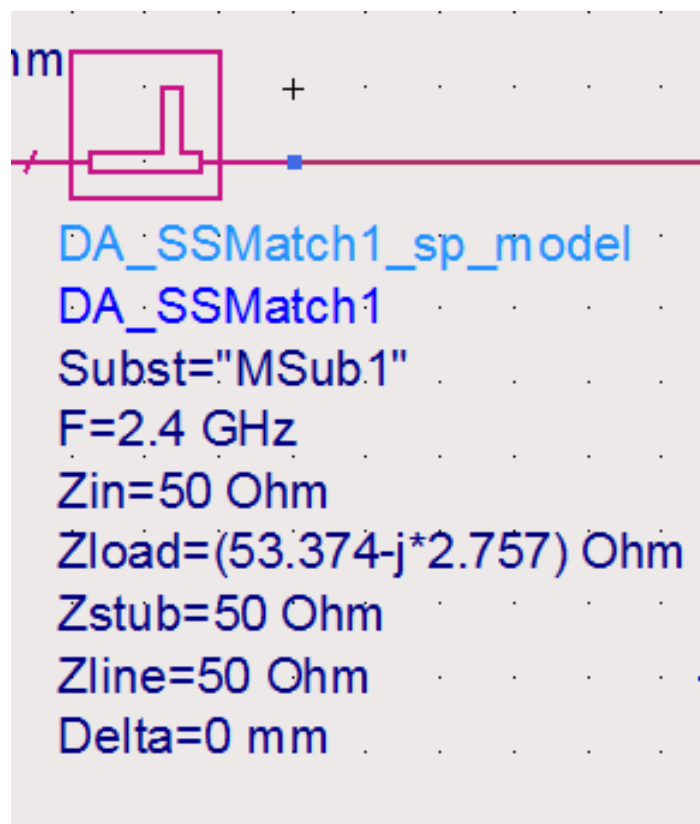


➤ 设置优化方式和优化目标，进行优化

 OPTIM	GOAL	GOAL	GOAL
Optim Optim 1 Optim Type=Random MaxIters=500 SaveAllTrials=no	Goal Optim Goal1 Expr="dB(S(1,1))" Sim InstanceName="SP1" Weight=1.0 IndepVar[1]="freq" LimitType[1]="LessThan" LimitMin[1]= LimitMax[1]=-15 Indep1Min[1]=2.39 GHz Indep1Max[1]=2.41 GHz	Goal Optim Goal2 Expr="dB(S(2,1))" Sim InstanceName="SP1" Weight=1.0 IndepVar[1]="freq" LimitType[1]="GreaterThan" LimitMin[1]=13 LimitMax[1]= Indep1Min[1]=2.39 GHz Indep1Max[1]=2.41 GHz	Goal Optim Goal3 Expr="dB(S(2,2))" Sim InstanceName="SP1" Weight=1.0 IndepVar[1]="freq" LimitType[1]="LessThan" LimitMin[1]= LimitMax[1]=-15 Indep1Min[1]=2.39 GHz Indep1Max[1]=2.41 GHz

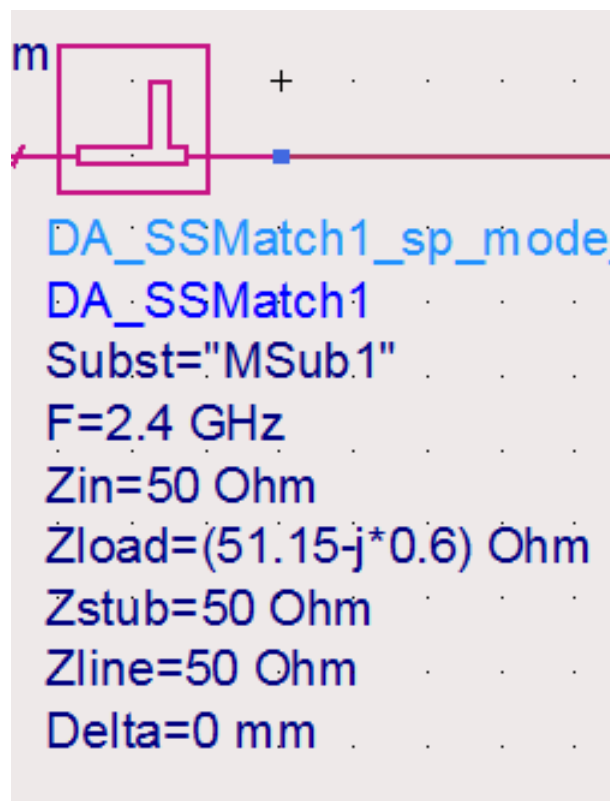


最小反射仿真结果(阻抗= $53.374 + j \cdot 2.757$)

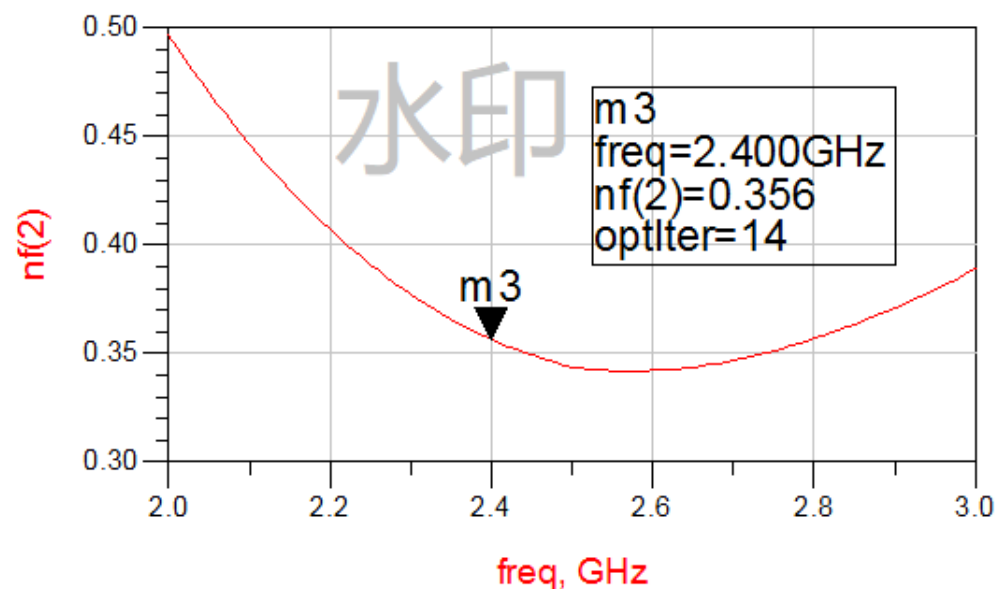
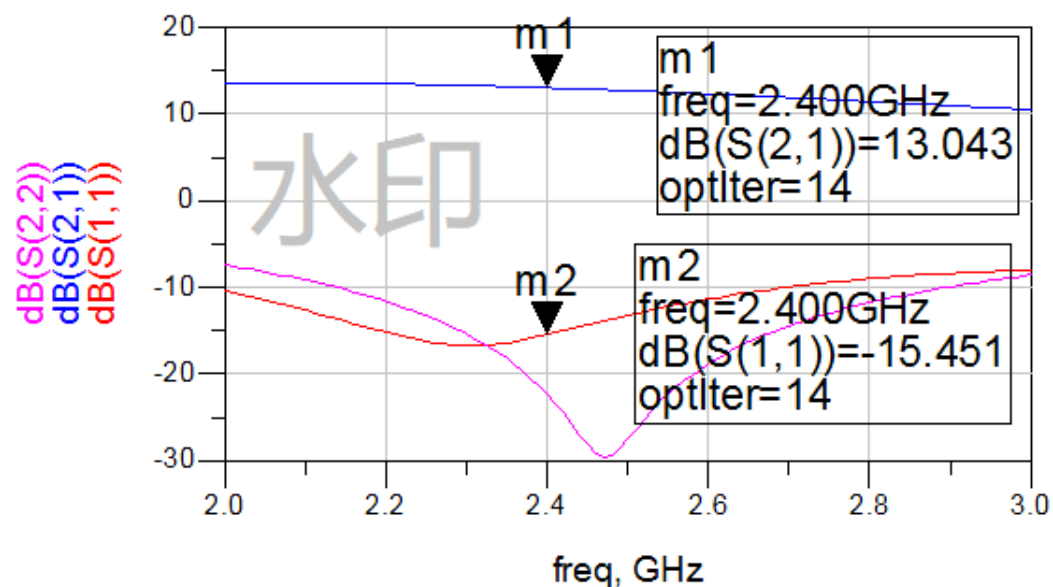




均衡仿真结果(阻抗= $51.15+j*0.6$)

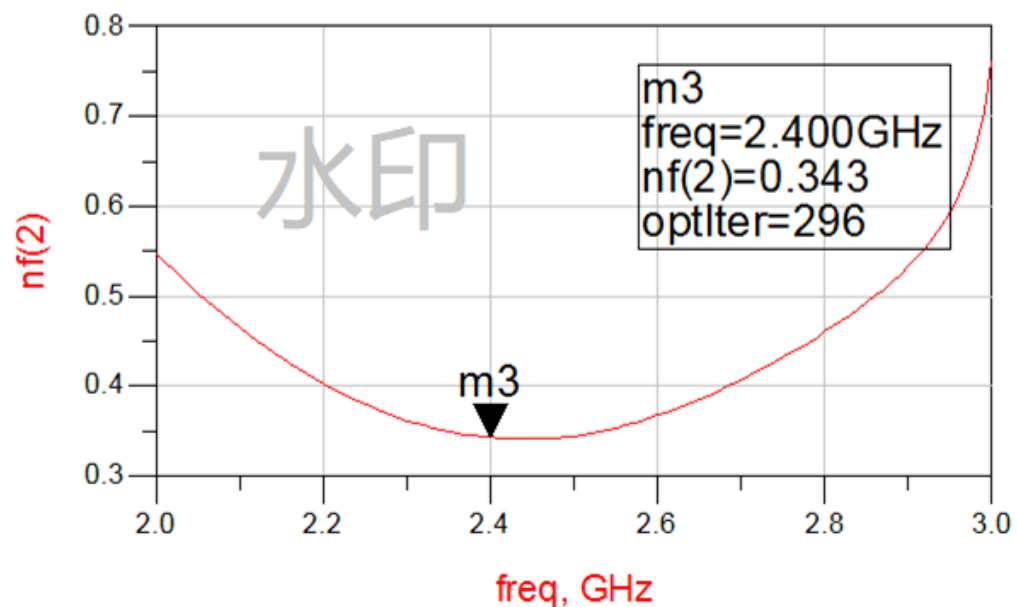
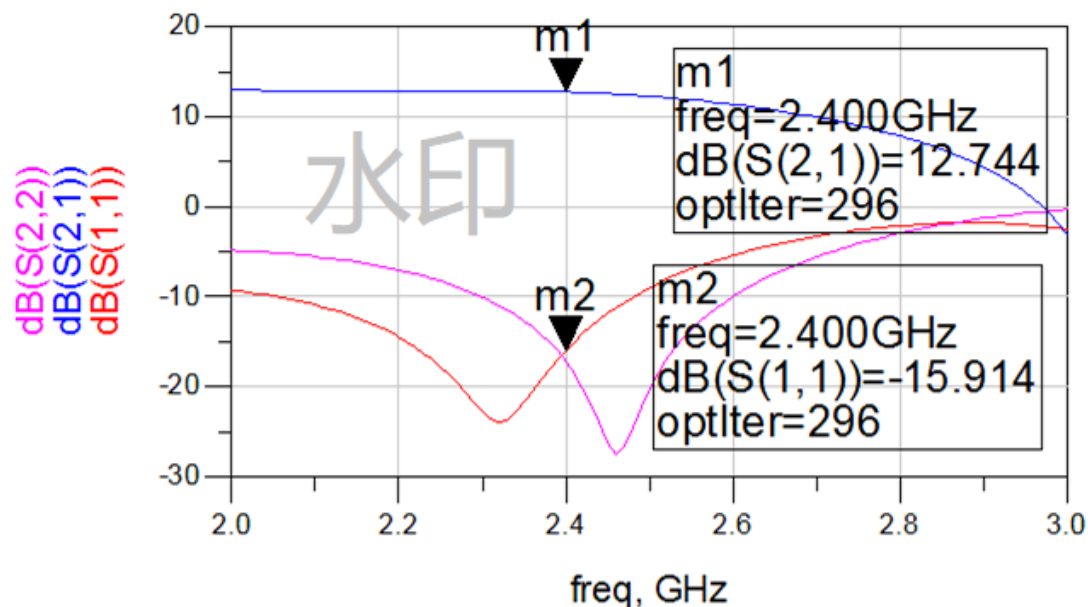
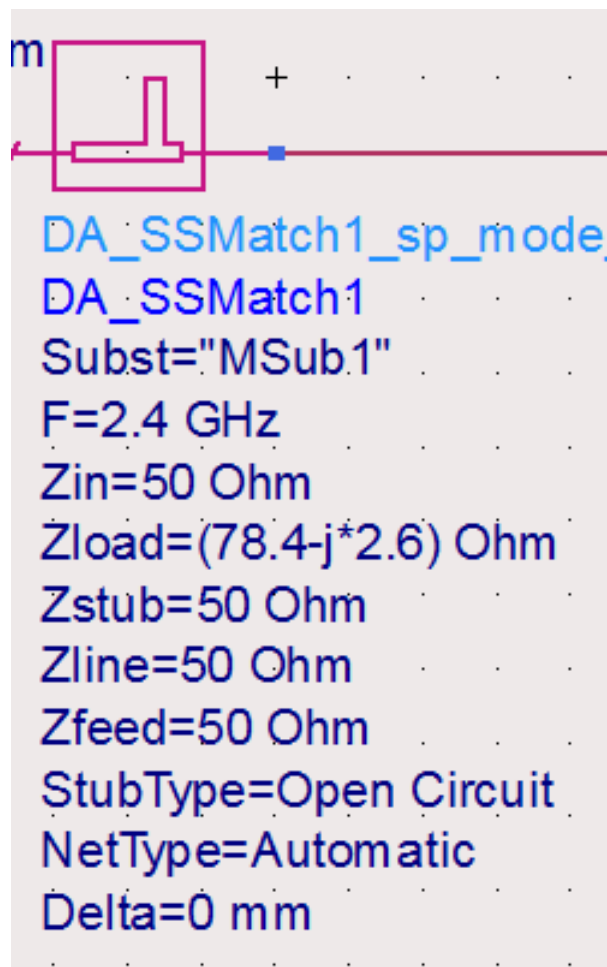


注意：
Zload改变时，需要先
对单支分节线进行重
新设计，然后再对输
出匹配进行优化。



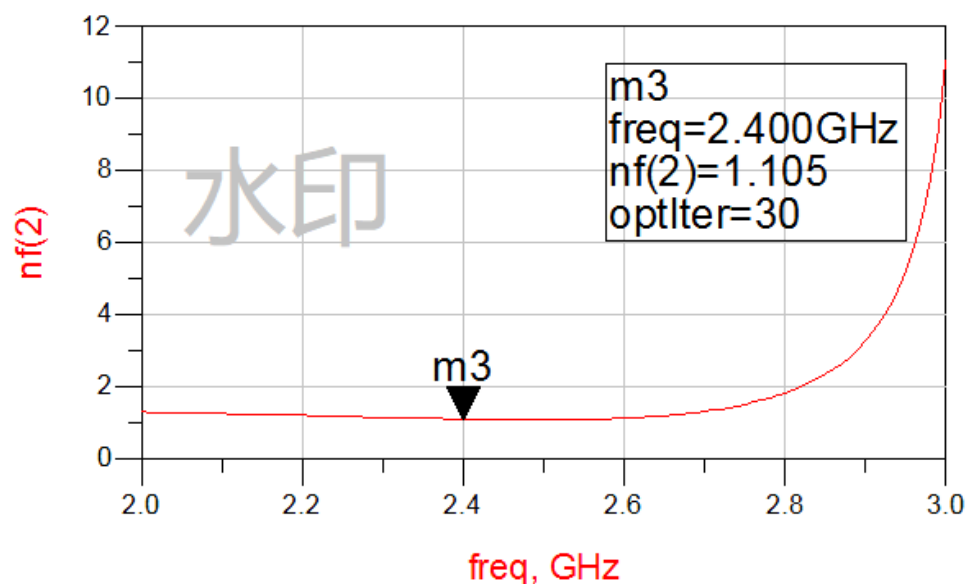
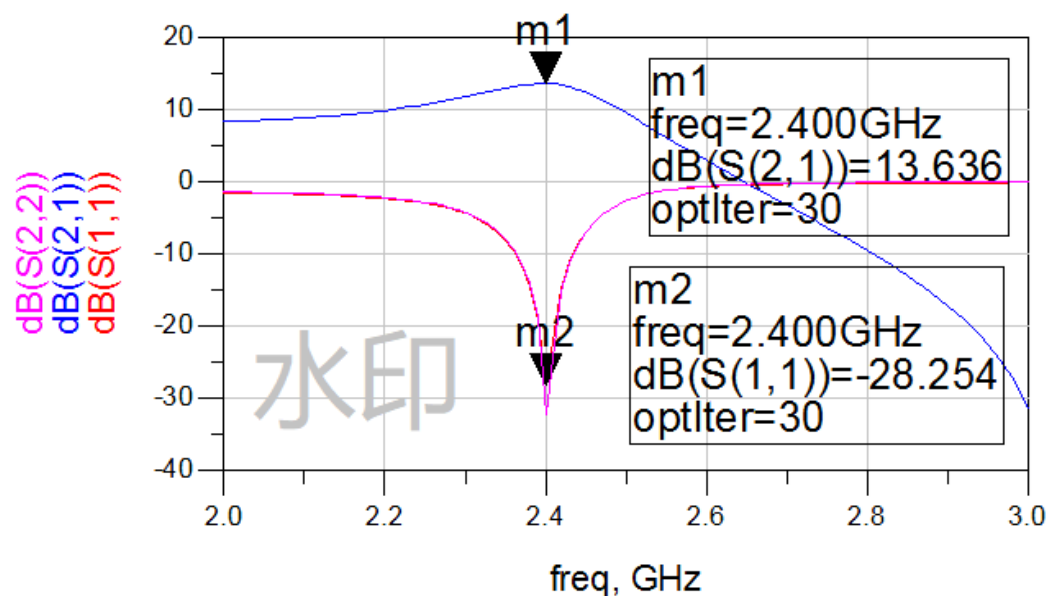
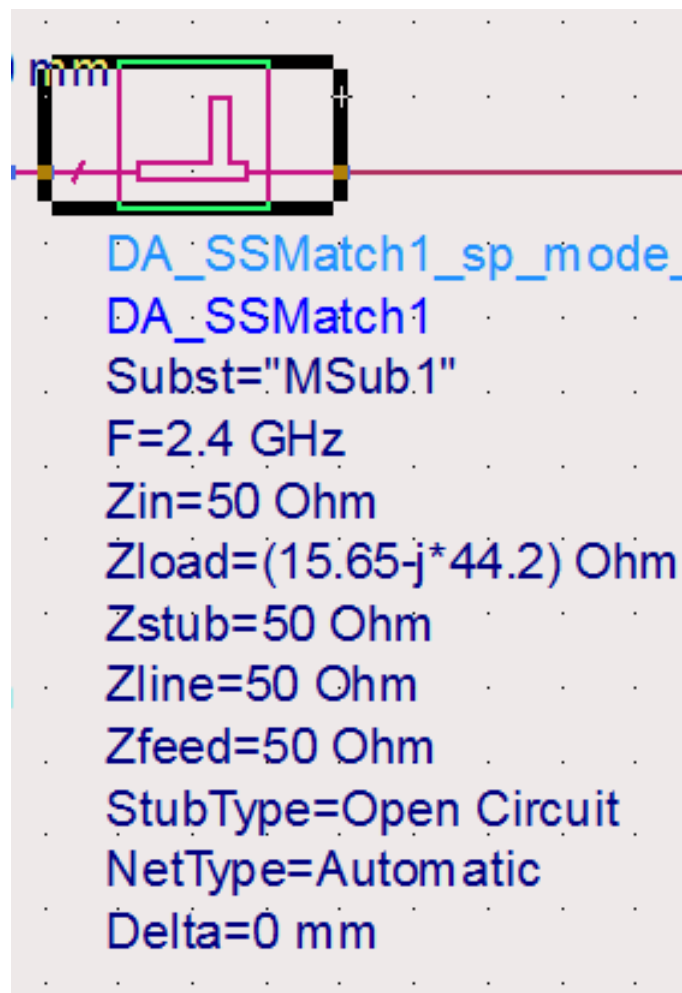


最小噪声仿真结果(阻抗= $78.4 + j*2.6$)





最大增益仿真结果(阻抗= $15.65+j*44.2$)





ADS仿真作业

在前面仿真的基础上

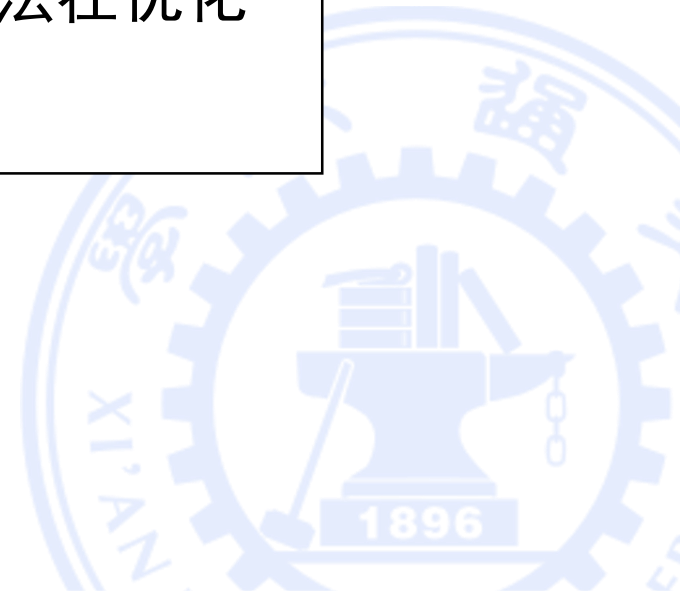
需仿真验证???

学号尾号为**单号**的同学完成**最大增益放大器的设计**，
指标：2.4GHz时，增益大于**13.6dB**，两个端口的反射小于-20dB。
要求：给出优化目标的设置，最后的原理图，以及仿真结果。

学号尾号为**双号**的同学完成**最小噪声放大器的设计**，
指标：2.4GHz时，噪声小于**0.34**，两个端口的反射小于-15dB。
要求：给出优化目标的设置，最后的原理图，以及仿真结果。



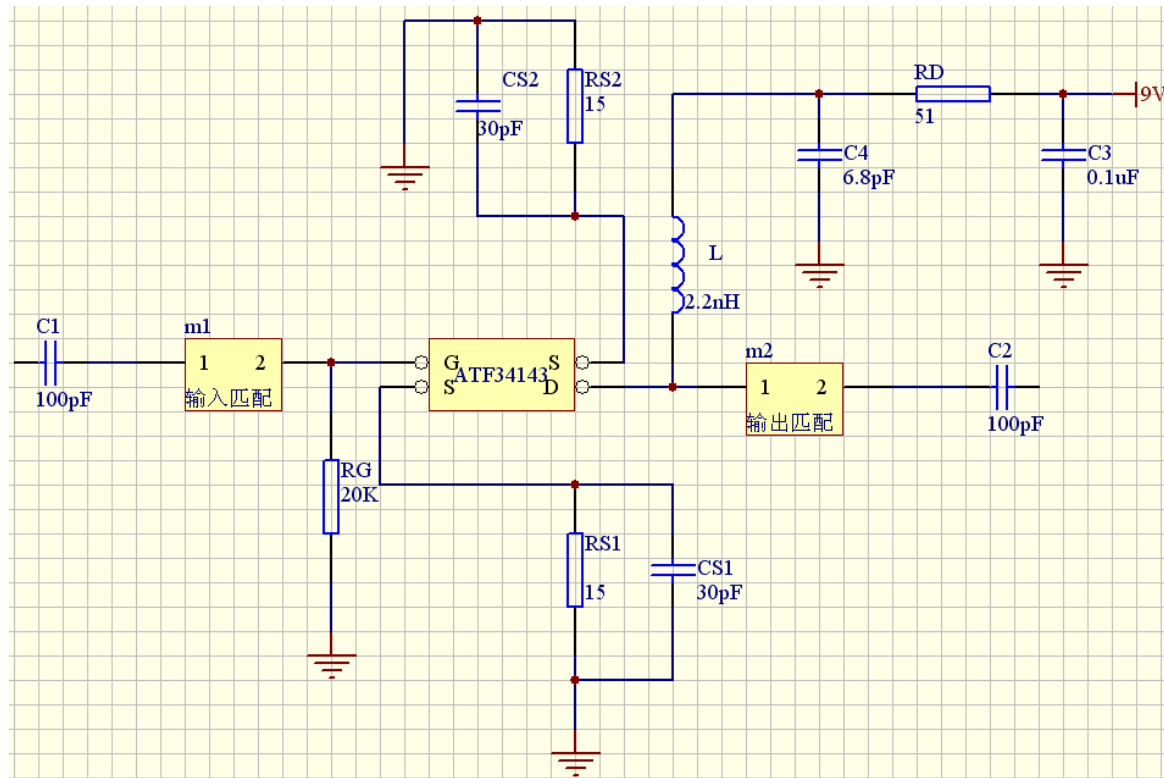
- 上述方法过于繁琐，比较适合刚接触ADS的初学者。
通过预先对sp模型进行仿真，可以获得匹配电路的大概指标，以便简化后期优化。
- 我们这里采用直接封装模型，设计和优化包括晶体管、偏置、输入匹配和负载（输出匹配）的完整电路。这种方法在优化上比较费事。

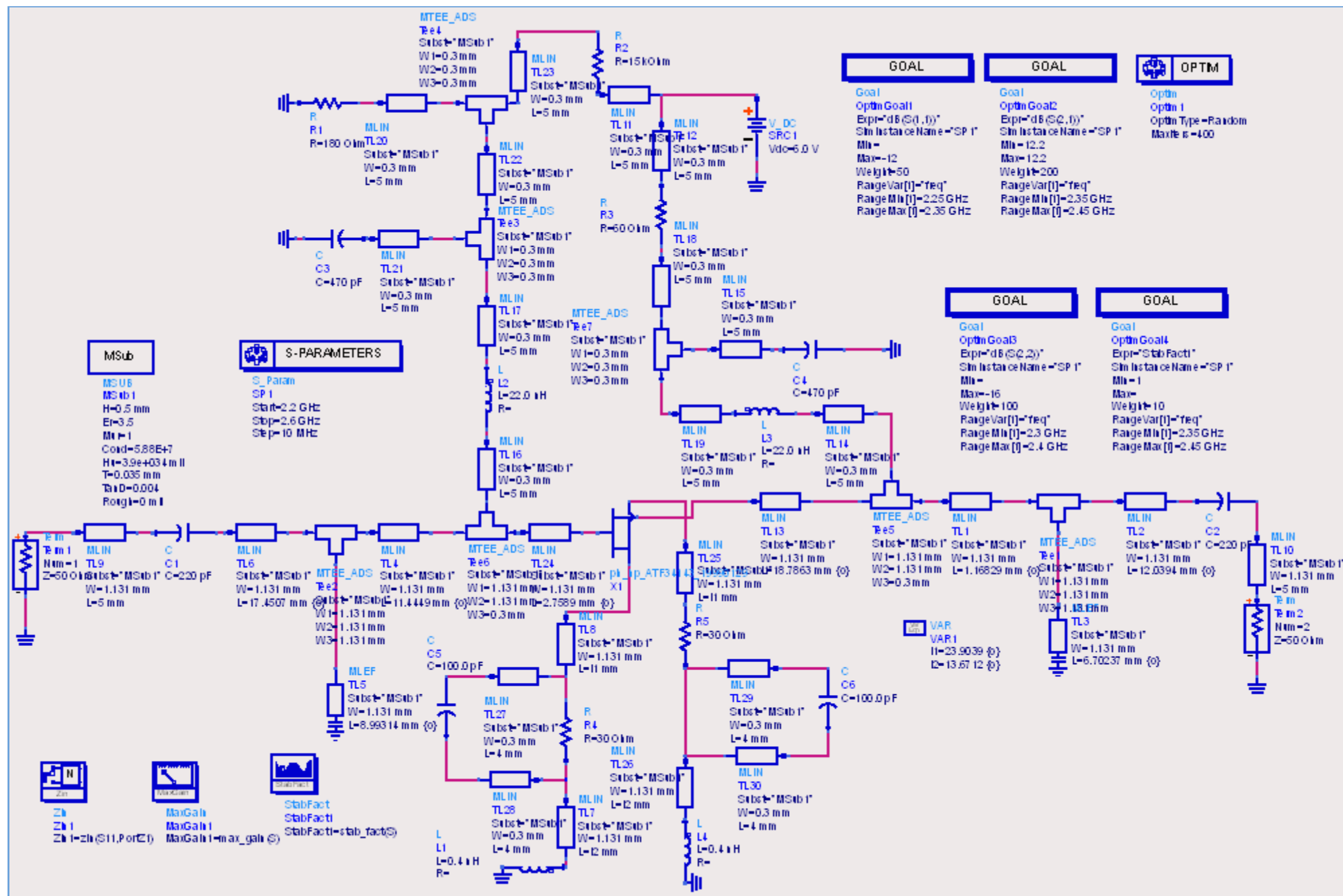




低噪放电路结构

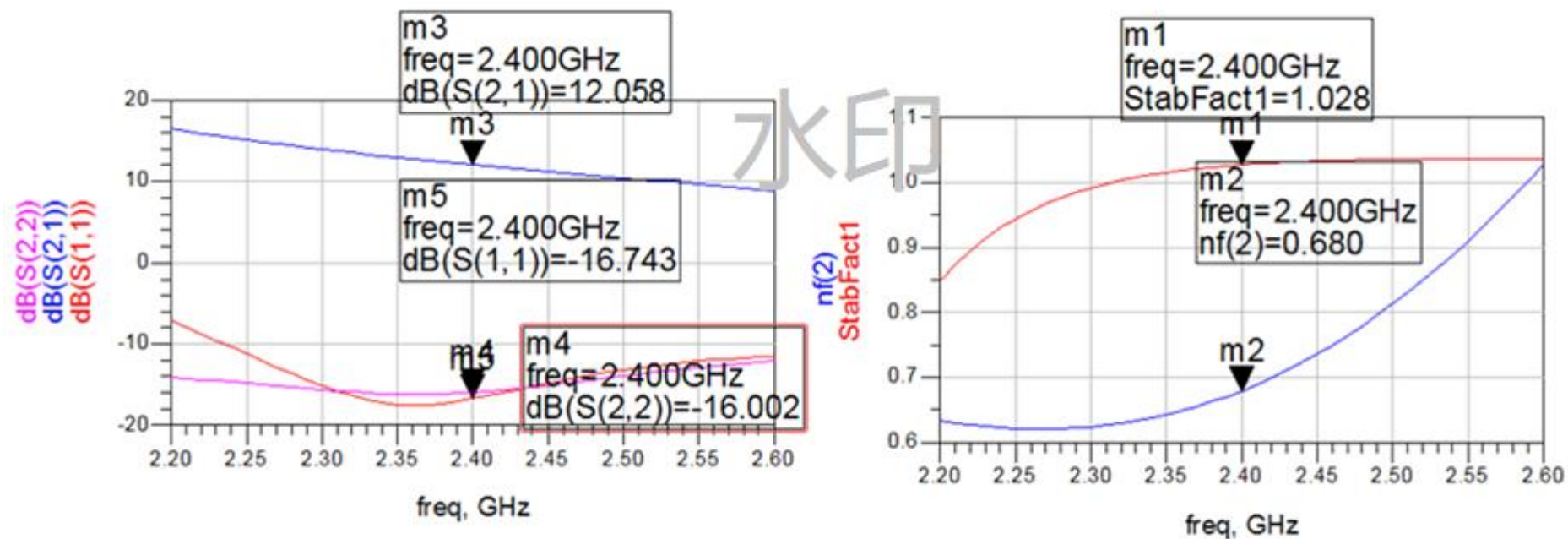
- 偏置电路采用自给偏压式。
- 匹配电路采用并联开路分支线的形式。
- ATF34143有两个S极，这是根据器件的实际封装。





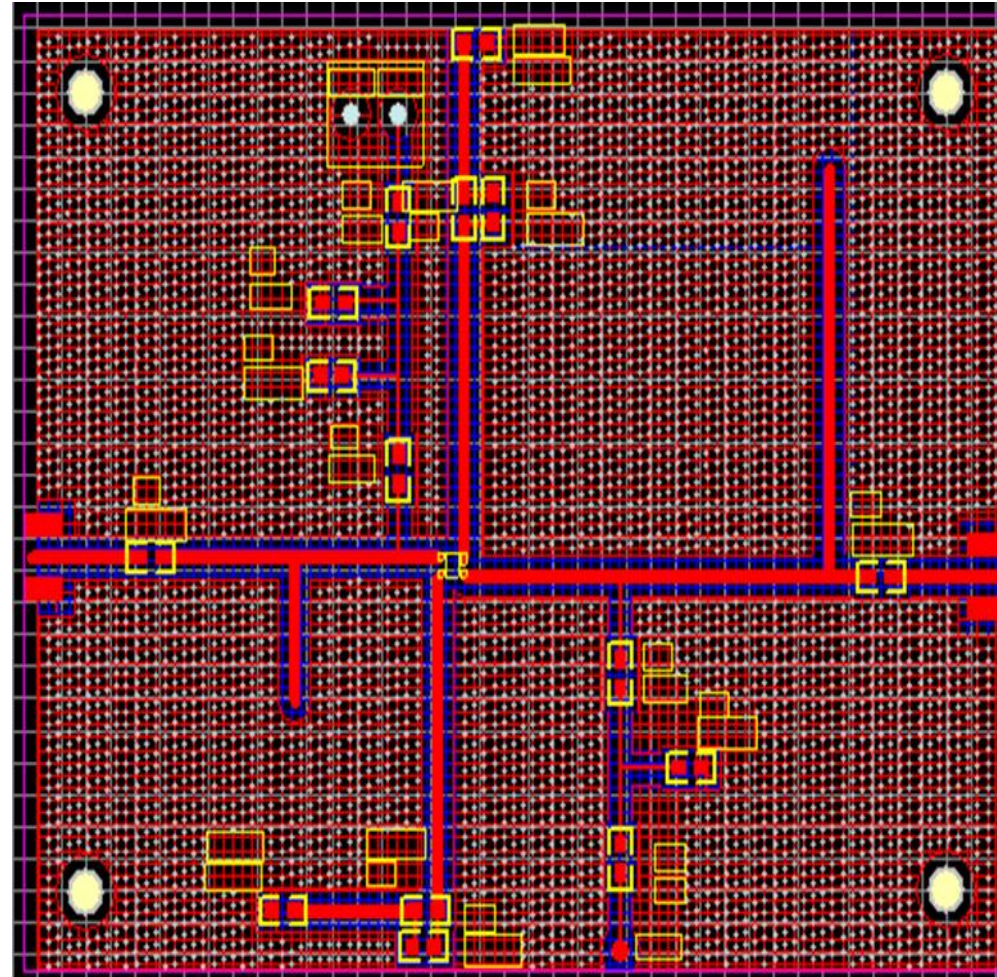


仿真结果





PCB版图为



根据仿真结果中微带线的长度等来画电路板。





四、微带天线制作与特性测量

- 根据第5次作业得到的数据，现场制作矩形贴片微带天线；
- 用矢量网络分析仪测量天线的驻波和阻抗(15点)，并确定中心频率。
- 验证自己所查的扩展带宽或实现多频的方法。

微带天线参数测量记录表

频率 (MHz)															
回损 (dB)															
Z_{in}															
驻波															
中心 频率															



四、微带天线制作与特性测量

(选做) AV4036B频谱分析仪 + AV1441信号源

- 选两个中心频率相近的微带天线。
- 将一个微带天线接信号源，设置如下：

频率 刚确定的中心频率

功率 -5dBm

- 将另一个微带天线接频谱分析仪，如下设置：

频率 刚确定的中心频率

频宽 50MHz

带宽

RBW: 1MHz

视频平均 【开】

幅度 参考电平 【-##dBm】

幅度刻度 【##dB】

使波形显示合适

调整**接收天线与发射天线的相对位置，相对角度**，观察记录频谱仪测量的信号峰值幅度，分析幅度变化的原因。

频标 峰值 频标→中心



五、思考题

1. 若放大器的稳定性 $k < 1$ ，可否同时实现输入、输出网络共轭匹配？
2. 放大器设计的具体步骤有哪些？
3. 如何给仿真设计加噪声？



六、本次实验报告格式

一、实验目的

二、放大器仿真设计

- (1) 给出优化目标的设置；
- (2) 给出原理图及其仿真结果，并对其结果进行分析。

三、微带天线测试结果及分析

- (1) 用于制作微带天线的各结构尺寸参数；
- (2) 数据表格；
- (3) 所制作天线的实物照片，驻波比的曲线照片；
- (4) 绘制天线回损的数据曲线；
- (5) 给出初始和调整中心频率后的曲线照片及中心频率值；
- (6) 验证增加带宽或实现多频的方法。

四、实验体会和建议

五、思考题